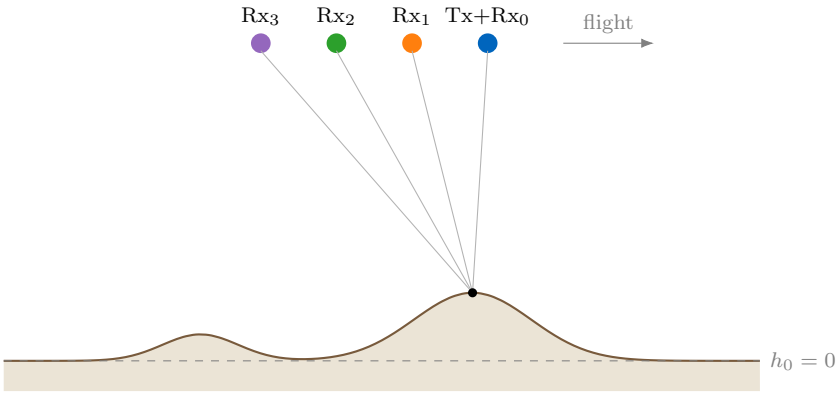
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Topography-Dependent Azimuth Reconstruction in Multistatic SAR Systems

Project report



Lucas Ayub Rodrigues Tuler

Instituto Tecnológico de Aeronáutica (ITA), Brazil

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Abstract

Multichannel and multistatic synthetic aperture radar (SAR) systems for high-resolution wide-swath (HRWS) imaging sample the azimuth signal of each receiver at a pulse repetition frequency (PRF) below the Doppler bandwidth and recover the unambiguous spectrum by combining the channels in an azimuth reconstruction. The reconstruction filters are derived for a reference geometry. When the receivers are separated by cross-track baselines, a scatterer at a height different from the reference height carries a channel-dependent phase, the aliased parts of the spectrum no longer cancel, and azimuth ambiguities appear.

This report presents a *joint* topography correction and azimuth reconstruction that removes these ambiguities using *only the system parameters and a digital elevation model (DEM)*, never the positions of the targets. The phase that the terrain adds to each channel is computed from the DEM and placed inside the model of the channels, at every position of the line; the focused line is then estimated by regularised least squares with conjugate gradients, applying the model and its adjoint with FFTs only. The method is developed and tested with a numerical simulator of a spaceborne formation ($\lambda = 0.25\,\mathrm{m}$, $r_0 = 766\,\mathrm{km}$, $N_{\mathrm{rx}} = 2$ to 8 receivers, cross-track baselines up to 500 m).

Over 1040 one-dimensional runs (52 scenes, $N_{\mathrm{rx}} = 2$–8) the median highest ambiguity drops from $-8.4\,\mathrm{dB}$ without correction to $-40.6\,\mathrm{dB}$ with the joint reconstruction (reference $-47.6\,\mathrm{dB}$). The two-dimensional extension, which applies the DEM at every pixel of a 3-D scene, reaches a median of $-44.4\,\mathrm{dB}$ over 640 runs (no correction $-13.9\,\mathrm{dB}$, reference $-47.6\,\mathrm{dB}$). On real terrain, 200 azimuth lines cut from the SRTM DEM of the Sierra Nevada, the median goes from $-8.3\,\mathrm{dB}$ to $-41.5\,\mathrm{dB}$, at the level of the reference ($-41.6\,\mathrm{dB}$). The method works best with three or more channels and cross-track baselines that are not equally spaced, which give the channels the diversity needed to separate the folded parts of the spectrum.

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Symbols and acronyms

Symbol	Meaning
λ	wavelength (0.25 m)
v	platform velocity
r_0	slant range of closest approach of the reference point
θ	look angle ($\approx 20°$)
N_{rx}	number of receive channels (one transmitter, N_{rx} receivers)
$b_{\mathrm{at},i}$, $b_{\mathrm{xt},i}$	along-track and cross-track baseline of receiver i relative to the transmitter
PRF	full (reconstructed) pulse repetition frequency, 2 000 Hz
$\mathrm{PRF}_{\mathrm{op}}$	PRF of each channel, $\mathrm{PRF}/N_{\mathrm{rx}}$
B_{a}	processed azimuth (Doppler) bandwidth, 1 235 Hz
$C_0, C_1, C_2, \Delta t$	coefficients of the bistatic minus monostatic path of a channel (Section 2.3)
$\Delta C_{0,i}(x)$	topographic residual of channel i: C_0 of a scatterer at height $h(x)$ minus C_0 of the reference point
X	ground distance between parcels folded onto the same Doppler bin, $X \approx r_0 \lambda \mathrm{PRF}_{\mathrm{op}}/(2v)$
$H_i(f)$	transfer function of channel i relative to the monostatic reference
$g(x)$	focused full-rate azimuth line (unknown of the joint reconstruction)
μ	Tikhonov regularisation parameter

Acronym	
CGLS	conjugate gradients for least squares
DEM	digital elevation model
DPCA	displaced phase centre antenna (uniform effective sampling)
HRWS	high-resolution wide-swath
IRF	impulse response function

1 Introduction

1.1 Context

A conventional SAR cannot improve its azimuth resolution and its swath width at the same time: a fine azimuth resolution needs a short antenna and therefore a wide Doppler bandwidth, which must be sampled with a high PRF, while a wide swath needs a long interval between pulses. Multichannel systems break this trade-off by receiving each pulse with several apertures displaced along track [1, 2]. Each channel is sampled at a PRF below the Doppler bandwidth and is therefore aliased, but the set of channels samples the azimuth signal at the full rate, and the unambiguous spectrum is recovered by an *azimuth reconstruction*: a bank of filters that combines the channels so that the aliased parts cancel.

Multistatic formations distribute the receivers over several platforms. They lift the limit set by the length of a single antenna and offer flexibility, but the baselines between platforms are no longer small [3, 4]. In particular, a *cross-track* baseline makes the bistatic range history of a scatterer depend on its height. The reconstruction filters are derived for a reference geometry (here: the reference point of the scene at height $h_0 = 0$); a scatterer at another height carries a residual phase that is different in each channel. The filters then no longer match the data, the aliased parts no longer cancel and azimuth ambiguities appear in the image.

The same kind of height-dependent phase is known from airborne motion compensation, where it is computed from an external DEM and removed position by position [5, 6]. This work builds on that idea: the phase that the terrain adds to each channel is known from the DEM and the system geometry, and the question is how to use it so that the multichannel reconstruction recovers an image free of ambiguities.

1.2 Objective

The goal is to remove the azimuth ambiguities caused by topography and cross-track baselines in multistatic SAR systems, using *only the system parameters and a DEM*, with no knowledge of where the scatterers are. The work answers two questions:

1. How large are the ambiguities caused by unaccounted topography, as a function of the terrain, the baselines and the number of channels?
2. How can the DEM be used inside the reconstruction so that the image reaches the quality of an ideal, unaliased reference?

1.3 Contributions

- A numerical simulator of the multistatic acquisition and of the multichannel reconstruction in one dimension (azimuth) and in two dimensions (azimuth and range, 3-D scenes), with along- and cross-track baselines and arbitrary terrain (Chapters 2 and 3).
- A joint topography correction and reconstruction that places the DEM phase of every position inside the model of the channels and inverts it over the whole line by regularised

least squares with conjugate gradients, using only FFTs (Chapter 4), and its 2-D extension (Chapter 5).

- Systematic sweeps over terrains, numbers of channels and baselines, and a validation on real terrain from the SRTM DEM (Chapter 6).

1.4 Outline

Chapter 2 introduces the acquisition geometry and the topographic residual ΔC_0. Chapter 3 reviews the multichannel reconstruction and explains how topography produces ambiguities. Chapter 4 presents the joint reconstruction and Chapter 5 its 2-D version. Chapter 6 collects the results and Chapter 7 concludes. The simulator, the methods and the scripts that produce every figure of this report are available at github.com/lucas-ayub/dlr-project.

2 Acquisition geometry and signal model

2.1 Formation and parameters

One platform transmits and receives (channel 0); the other $N_{\mathrm{rx}} - 1$ receivers fly behind it. Receiver i is displaced by $b_{\mathrm{at},i}$ along track and by $b_{\mathrm{xt},i}$ across track (horizontally, towards the scene) relative to the transmitter (Figure 2.1). The along-track baselines satisfy the displaced phase centre (DPCA) condition (Figure 2.2), so that the effective phase centres of the N_{rx} channels are uniformly spaced and the multichannel sampling is uniform at the full PRF; the study is about the cross-track baselines, which are drawn at random in $[0, {b_{\mathrm{xt}}}^{\max}]$ (with the largest equal to ${b_{\mathrm{xt}}}^{\max}$) or taken from fixed sets. Channel 0 is always the transmitter ($b_{\mathrm{at}} = b_{\mathrm{xt}} = 0$). The parameters are listed in Table 2.1.

Table 2.1: Parameters of the simulated system.

Wavelength λ	0.25 m
Platform altitude H	720 km
Slant range of the reference point r_0	766.2 km
Look angle θ	20°
Platform velocity / effective velocity	7 688.5 m/s / 7 408.5 m/s
Antenna length	12 m
Processed Doppler bandwidth B_{a}	1 235 Hz
Full PRF	2 000 Hz
Channel PRF $\mathrm{PRF}_{\mathrm{op}} = \mathrm{PRF}/N_{\mathrm{rx}}$	1 000 Hz to 250 Hz ($N_{\mathrm{rx}} = 2$ to 8)
DPCA along-track spacing $2v/\mathrm{PRF}$	7.69 m
Cross-track baselines ${b_{\mathrm{xt}}}^{\max}$	100 m, 300 m and 500 m, and fixed sets

2.2 Signal model

The simulator works with point targets. In the 1-D study the targets lie on a terrain profile $h(x)$ along azimuth; each target is placed on the iso-range surface of the reference point (same monostatic slant range r_0), so that only its height changes the bistatic geometry. The reference point is at $x = 0$, $h = 0$. In the 2-D study (Chapter 5) the terrain is a surface $h(x, y)$ and the data are range compressed. The azimuth signal of channel i is

$$s_i(t) = \sum_p a_p \, w(t - t_p) \, \exp\left(-\mathrm{j} \frac{2\pi}{\lambda} \left[r_{\mathrm{T},p}(t) + r_{\mathrm{R},i,p}(t) \right] \right), \tag{2.1}$$

with a_p the complex amplitude of scatterer p, w the antenna pattern and $r_{\mathrm{T}} + r_{\mathrm{R},i}$ the bistatic path (Figure 2.1).

2.3 Channel transfer functions and the coefficients C_0, C_1, C_2, Δt

The reconstruction (Chapter 3) needs, for every channel i, its transfer function $H_i(f)$: the factor by which the Doppler spectrum of channel i differs from the spectrum the monostatic system (transmitter and receiver at the transmitter) would record for the same scatterer. This

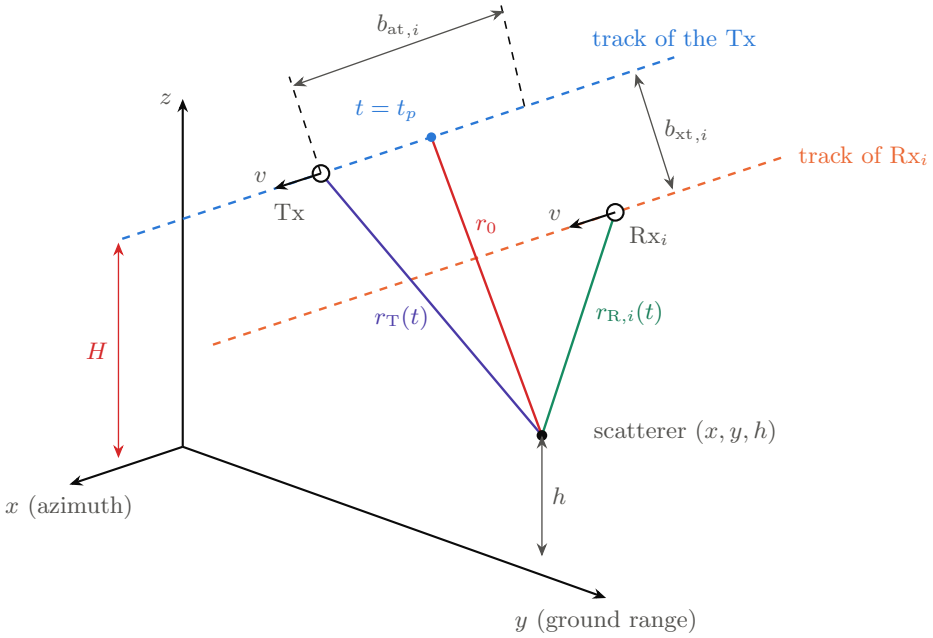


Figure 2.1: Geometry of one channel. The transmitter (Tx) and receiver i (Rx$_i$) fly on parallel tracks at the same altitude H, separated by $b_{\mathrm{at},i}$ along track and $b_{\mathrm{xt},i}$ across track. The echo of channel i travels $r_{\mathrm{T}}(t) + r_{\mathrm{R},i}(t)$. The time of closest approach t_p and the range r_0 fix the position of the scatterer on the line; its height h is what the topography adds. Not to scale.

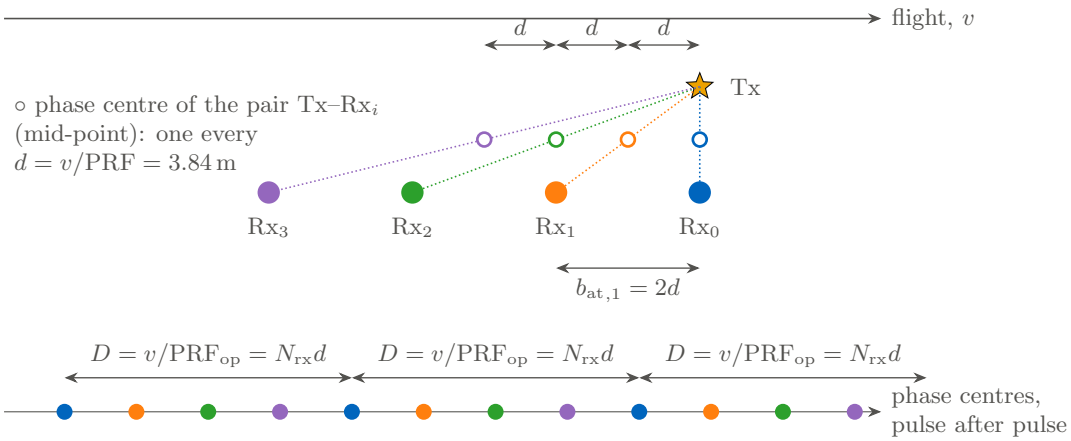


Figure 2.2: DPCA arrangement seen from above ($N_{\mathrm{rx}} = 4$). Receiver i trails the transmitter by $b_{\mathrm{at},i} = 2id$, so the phase centre of the pair (the mid-point) trails it by id. At every pulse the four phase centres occupy four equally spaced positions; between pulses the platform moves $D = N_{\mathrm{rx}}d$ and the pattern repeats. Together the channels sample the ground every d, like a single channel at the full PRF.

section derives H_i from the range histories of the reference point; the derivation follows the Doppler-domain approach of [3, 4].

Step 1: two parabolas (time domain). Let $M(t) = 2r_{\mathrm{T}}(t)$ be the monostatic path and $B_i(t) = r_{\mathrm{T}}(t) + r_{\mathrm{R},i}(t)$ the bistatic path of channel i, both for the reference point. Over the aperture each is well described by a parabola around its own minimum:

$$M(t) \approx M_{\min} + m_2\,(t - t_M)^2, \qquad B_i(t) \approx B_{i,\min} + b_{2,i}\,(t - t_{B,i})^2, \tag{2.2}$$

where $M_{\min}$, $B_{i,\min}$ are the smallest path lengths [m], t_M, $t_{B,i}$ the times at which they occur [s] and m_2, $b_{2,i}$ the curvatures [m/s^2]. The corresponding azimuth signals are $s_M(t) = w(t)\exp[-\mathrm{j}2\pi M(t)/\lambda]$ and $s_i(t) = w(t)\exp[-\mathrm{j}2\pi B_i(t)/\lambda]$.

Step 2: from time to Doppler (stationary phase). For a signal $\exp[-\mathrm{j}2\pi P(t)/\lambda]$ with a parabolic path $P(t) = P_{\min} + p_2(t - t_P)^2$, the instantaneous frequency is

$$f(t) = -\frac{1}{\lambda}\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{2p_2}{\lambda}(t - t_P) \quad \Longrightarrow \quad t_f = t_P - \frac{\lambda f}{2p_2}, \tag{2.3}$$

so every Doppler frequency f is produced at one time t_f (the same relation as the ruler (3.3) of Chapter 3). By the principle of stationary phase the phase of the spectrum at f is the phase of the signal at t_f minus $2\pi f t_f$. With $P(t_f) = P_{\min} + \lambda^2 f^2/(4p_2)$,

$$\Psi_P(f) = -\frac{2\pi}{\lambda}P(t_f) - 2\pi f t_f = -2\pi\Big[\frac{P_{\min}}{\lambda} + t_P\,f - \frac{\lambda f^2}{4p_2}\Big]. \tag{2.4}$$

The amplitude of the spectrum is set by the antenna pattern and is the same for the two paths.

Step 3: the ratio of the two spectra (Doppler domain). Applying (2.4) to B_i and to M and taking the difference of the phases gives the transfer function of channel i:

$$H_i(f) = \mathrm{e}^{\mathrm{j}[\Psi_{B_i}(f) - \Psi_M(f)]} = \exp\Big(-\mathrm{j}2\pi\Big[\frac{C_{0,i}}{\lambda} + (\Delta t_i - C_{1,i})\,f + C_{2,i}\,\lambda f^2\Big]\Big), \tag{2.5}$$

with

$$C_{0,i} = B_{i,\min} - M_{\min}, \qquad \Delta t_i - C_{1,i} = t_{B,i} - t_M, \qquad C_{2,i} = \frac{1}{4m_2} - \frac{1}{4b_{2,i}}. \tag{2.6}$$

In the simulator the two paths are sampled at the full PRF: Δt_i is the time between the samples where M and B_i are smallest and $C_{1,i}$ the sub-sample correction given by the linear terms of the two fitted parabolas, so that $\Delta t_i - C_{1,i}$ is the time between the two minima.

Physical meaning. Each coefficient corresponds to one effect of the bistatic geometry:

- $C_{0,i}$ [m], the difference of the shortest paths, is a *constant phase* $2\pi C_{0,i}/\lambda$. It is set by the cross-track baseline, and it is the coefficient that changes with the height of the scatterer (Section 2.4).
- $\Delta t_i - C_{1,i}$ [s], the delay between the two minima, is a *linear phase* in Doppler, i.e. a shift in time. It is set by the along-track baseline ($\approx b_{\mathrm{at},i}/(2v)$) and is what makes the channels sample the azimuth signal at displaced positions (DPCA).
- $C_{2,i}$ [s^2/m], the difference of the inverse curvatures, is a *quadratic phase*: a small residual azimuth chirp.

For a scatterer other than the reference point the same derivation holds with its own paths; what changes, to first order, is $C_{0,i}$ (Section 2.4).

Figure 2.3 shows the four coefficients of the reference point. C_0 is the difference of the closest-approach distances of the receiver and of the transmitter, $\approx -b_{\mathrm{xt}} \sin\theta$ ($-171\,\mathrm{m}$ for $b_{\mathrm{xt}} = 500\,\mathrm{m}$), and does not depend on b_{at}; $\Delta t = b_{\mathrm{at}}/(2v)$ (0.5 ms, 1.0 ms and 1.5 ms for the three DPCA receivers) and does not depend on b_{xt}; C_1 is below $0.2\,\mu\mathrm{s}$ and C_2 below $4 \times 10^{-7}\,\mathrm{s}^2/\mathrm{m}$, which at the edge of the processed band, $f = B_{\mathrm{a}}/2$, give phases of $2\pi C_1 f < 1 \times 10^{-3}$ rad and $2\pi C_2 \lambda f^2 \approx 0.2$ rad. Figure 2.4 checks the parabolic model (2.2) for one target: the bistatic minus the monostatic path varies by a few millimetres over the aperture (5.6 mm for $b_{\mathrm{xt}} = 300\,\mathrm{m}$) and the parabola reproduces it within 10^{-4} mm.

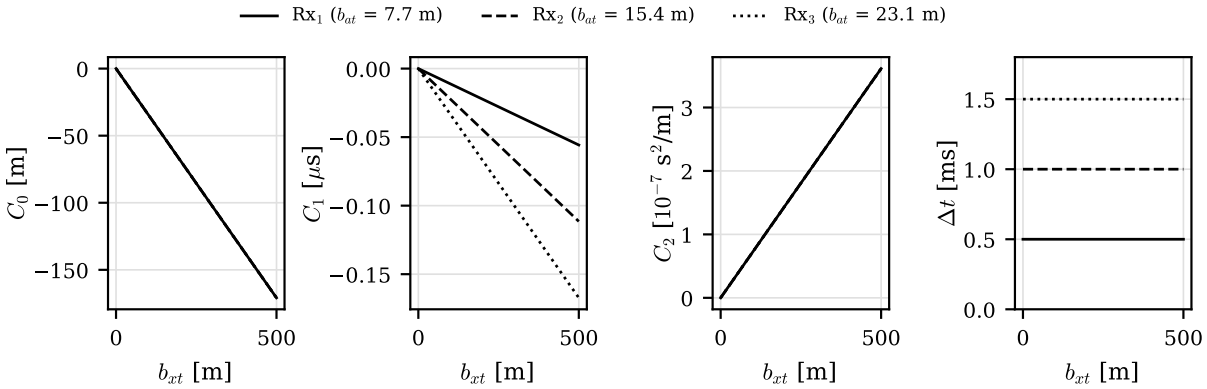


Figure 2.3: C_0, C_1, C_2 and Δt of the reference point for the receivers 1, 2 and 3 of the DPCA formation ($b_{\mathrm{at}} = 7.7\,\mathrm{m}$, 15.4 m and 23.1 m), as a function of the cross-track baseline b_{xt} (the same for the three receivers). The three curves of C_0 and of C_2 coincide.

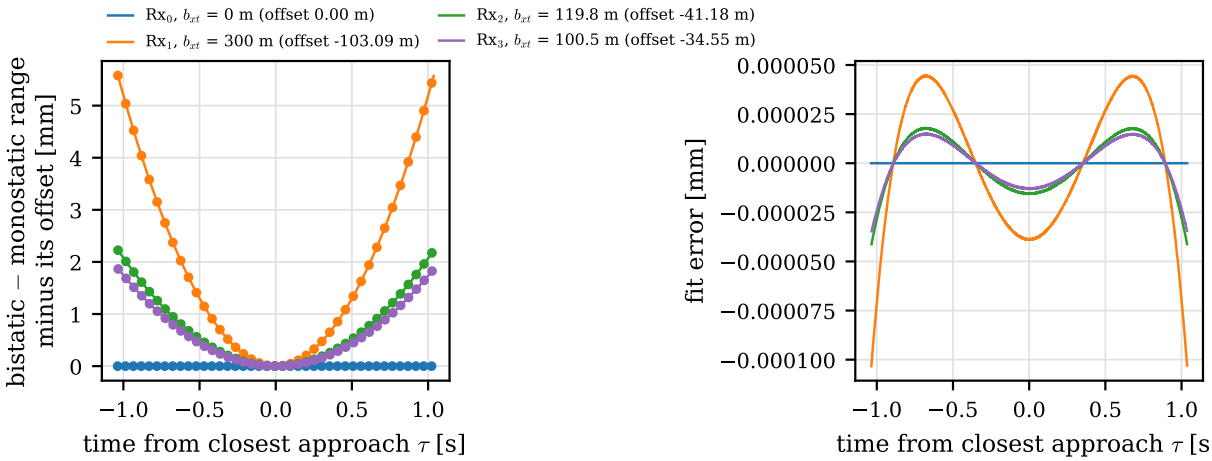


Figure 2.4: One target at $x = 0$, $h = 500\,\mathrm{m}$, four channels with $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m. Left: bistatic minus monostatic path of every channel against the time from closest approach, minus its constant offset (C_0, given in the legend); dots: simulated geometry; lines: the fitted parabola. Right: difference between the simulated curve and the parabola.

2.4 The topographic residual ΔC_0

For a scatterer at height h the coefficient C_0 of channel i differs from the one of the reference point. The difference is the *topographic residual*

$$\Delta C_{0,i}(h) = C_{0,i}(\text{scatterer}) - C_{0,i}(\text{reference point}) \approx -\frac{b_{\text{xt},i}\, h}{r_0 \tan\theta}, \qquad \varphi_i(h) = \frac{2\pi}{\lambda}\Delta C_{0,i}(h), \quad (2.7)$$

i.e. the interferometric phase of the cross-track baseline; it vanishes for the transmitter channel ($b_{\text{xt}} = 0$), since the targets are on the iso-range surface. The simulator computes $\Delta C_{0,i}$ exactly from the range histories; the approximation (2.7) agrees within about 2 %. For a scatterer at 500 m and $b_{\text{xt}} = [63.7, 27.0, 4.1, 1.7]$ m the residuals are $[-113.9, -48.2, -7.3, -3.0]$ mm, i.e. up to half a wavelength. The other coefficients change much less with height: the topography acts as a channel-dependent *constant* phase over the aperture of each scatterer, a fact used by the joint reconstruction (Chapter 4). Figure 2.5 quantifies this: up to 3 km of height and $b_{\text{xt}} = 500$ m, ΔC_0 follows (2.7) within a few per cent, and at the edge of the processed band the phase produced by ΔC_0 (about 80 rad at 3 km with $b_{\text{xt}} = 300$ m) is four orders of magnitude larger than the one produced by ΔC_2 (4×10^{-3} rad) and seven larger than the one produced by ΔC_1 (4×10^{-6} rad); Δt does not change with height.

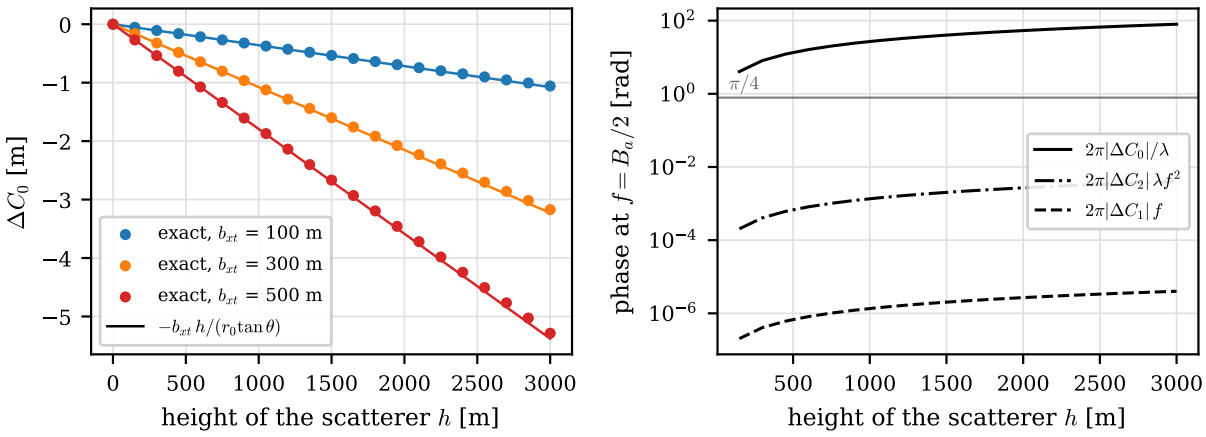


Figure 2.5: Change of the coefficients of receiver 1 when the scatterer is at height h instead of the reference point. Left: ΔC_0 (dots: exact, from the range histories; lines: (2.7)) for three cross-track baselines. Right ($b_{\text{xt}} = 300$ m): phase produced by each term of (2.5) at the edge of the processed band, $f = B_{\text{a}}/2$, on a logarithmic scale.

Figure 2.6 shows where (2.7) comes from. Moving from the reference point to a scatterer at height h on the same range arc of the transmitter is a displacement $s = h/\sin\theta$ perpendicular to the line of sight of the transmitter, which does not change its path. The line of sight of receiver i is rotated by $\Delta\theta = b_{\text{xt},i}\cos\theta/r_0$, so its path changes by $s\,\Delta\theta = b_{\text{xt},i}h/(r_0\tan\theta)$.

The sensitivity is $\partial\varphi_i/\partial h = 2\pi b_{\text{xt},i}/(\lambda r_0 \tan\theta) \approx 9.0 \times 10^{-5}\, b_{\text{xt},i}$ rad/m, which also sets the tolerance to DEM errors (Section 6.3).

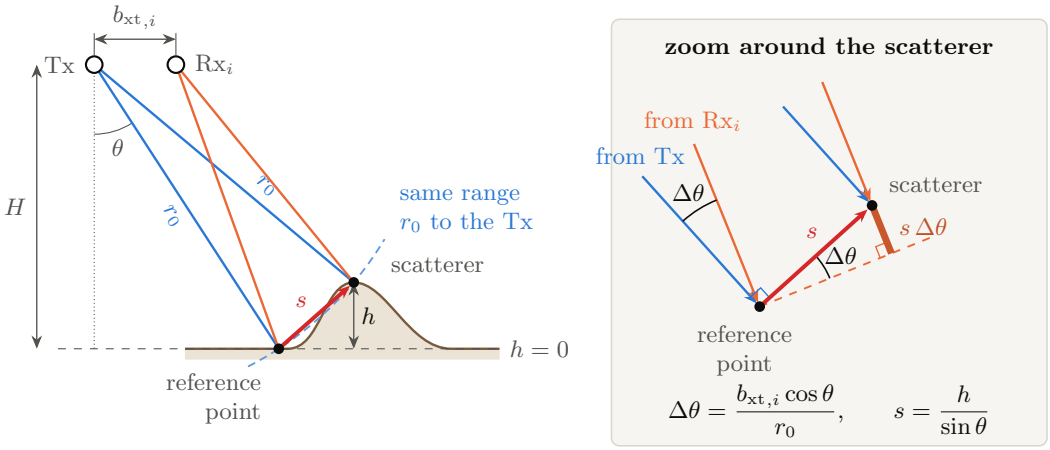


Figure 2.6: Left: plane perpendicular to the track (not to scale). The reference point and the scatterer are at the same range r_0 from the transmitter, at different heights: climbing h along the arc is a displacement $s = h/\sin\theta$. Right: zoom around the scatterer. Since $s \ll r_0$ the lines of sight of each platform to the two points are parallel; s is perpendicular to the line of sight of the transmitter, but the one of the receiver is rotated by $\Delta\theta$, and its path becomes $s\,\Delta\theta$ shorter (thick orange segment).

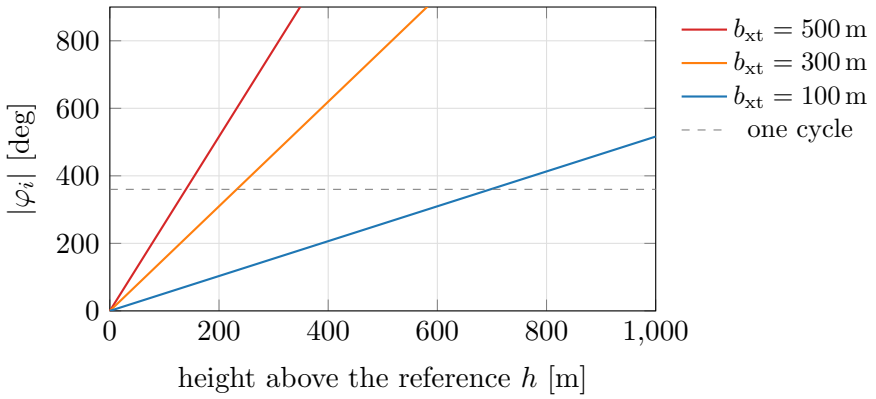


Figure 2.7: Phase of the topographic residual (2.7) against height. With $b_{\mathrm{xt}} = 300\,\mathrm{m}$ the sensitivity is 1.55° per metre of height: a 230 m hill already produces a full cycle.

3 Multichannel azimuth reconstruction and topography

3.1 Aliased channels and folded parcels

Each channel is sampled at $\mathrm{PRF}_{\mathrm{op}} = \mathrm{PRF}/N_{\mathrm{rx}}$, below the Doppler bandwidth $B_{\mathrm{a}} = 1\,235\,\mathrm{Hz}$. In the Doppler domain of the channel, every frequency $\phi \in [-\mathrm{PRF}_{\mathrm{op}}/2, \mathrm{PRF}_{\mathrm{op}}/2)$ is the sum of N_{rx} *parcels* of the full-rate spectrum U, taken at

$$f_q = \phi + q\,\mathrm{PRF}_{\mathrm{op}} - \tfrac{N_{\mathrm{rx}}-1}{2}\mathrm{PRF}_{\mathrm{op}}, \qquad q = 0, \dots, N_{\mathrm{rx}} - 1, \tag{3.1}$$

each weighted by the transfer function H_i of the channel relative to the monostatic reference:

$$S_i(\phi) = \sum_q H_i(f_q)\, U(f_q) \quad \Longleftrightarrow \quad \boldsymbol{s}(\phi) = \mathbf{H}(\phi)\, \boldsymbol{u}(\phi), \tag{3.2}$$

with $\boldsymbol{s}$ the N_{rx} channel spectra and $\boldsymbol{u}$ the N_{rx} parcels. The standard reconstruction [1, 2] inverts this square system at every ϕ, $\hat{\boldsymbol{u}} = \mathbf{P}(\phi)\boldsymbol{s}(\phi)$ with $\mathbf{P} = \mathbf{H}^{-1}$, and places the parcels at their true Doppler frequency to obtain the full-rate signal. The entries of $\mathbf{H}$ are the transfer functions (2.5) of the reference point, $H_{iq}(\phi) = H_i(f_q)$.

Through the Doppler–angle relation, a Doppler frequency f seen from the platform at azimuth position x_c corresponds to the ground position

$$x(f) = x_c + r_0 \tan\!\big(\arcsin(\lambda f/2v)\big) \approx x_c + \frac{\lambda r_0}{2v}\, f, \tag{3.3}$$

which we call the *ruler*. Parcels one $\mathrm{PRF}_{\mathrm{op}}$ apart therefore come from ground positions

$$X \approx \frac{\lambda r_0\, \mathrm{PRF}_{\mathrm{op}}}{2v} \qquad (6.23\,\mathrm{km} \text{ for } \mathrm{PRF}_{\mathrm{op}} = 500\,\mathrm{Hz}), \tag{3.4}$$

apart, i.e. 0.81 s of azimuth time. This distance is the key length of the problem: one bin of an aliased channel mixes the echoes of the terrain at x, $x \pm X$, $x \pm 2X$, … (Figure 3.1).

Figure 3.2 shows the two sides of this for one target on flat terrain and four channels. A single channel at $\mathrm{PRF}_{\mathrm{op}} = 500\,\mathrm{Hz}$ sees the processed band $B_{\mathrm{a}} = 1\,235\,\mathrm{Hz}$ folded two to three times onto its own band; the standard reconstruction combines the four channels and recovers the full-rate spectrum of the reference.

3.2 How topography creates ambiguities

For a scatterer at height h, channel i carries the extra constant phase $\mathrm{e}^{-\mathrm{j}\varphi_i}$ of (2.7), so the data are $\boldsymbol{s} = \mathbf{D}\mathbf{H}\boldsymbol{u}$ with $\mathbf{D} = \mathrm{diag}(\mathrm{e}^{-\mathrm{j}\varphi_i})$, and the standard reconstruction returns

$$\hat{\boldsymbol{u}} = \mathbf{H}^{-1}\mathbf{D}\,\mathbf{H}\,\boldsymbol{u} = \mathbf{L}\,\boldsymbol{u}. \tag{3.5}$$

In the DPCA configuration $\mathbf{H}$ is, up to diagonal factors, a discrete Fourier matrix over the channels, and the leakage matrix becomes Toeplitz,

$$L_{q'q} = c_{q'-q}, \qquad c_m = \frac{1}{N_{\mathrm{rx}}} \sum_{i=0}^{N_{\mathrm{rx}}-1} \exp\!\Big[\mathrm{j}\Big(\frac{2\pi i m}{N_{\mathrm{rx}}} - \varphi_i\Big)\Big]. \tag{3.6}$$

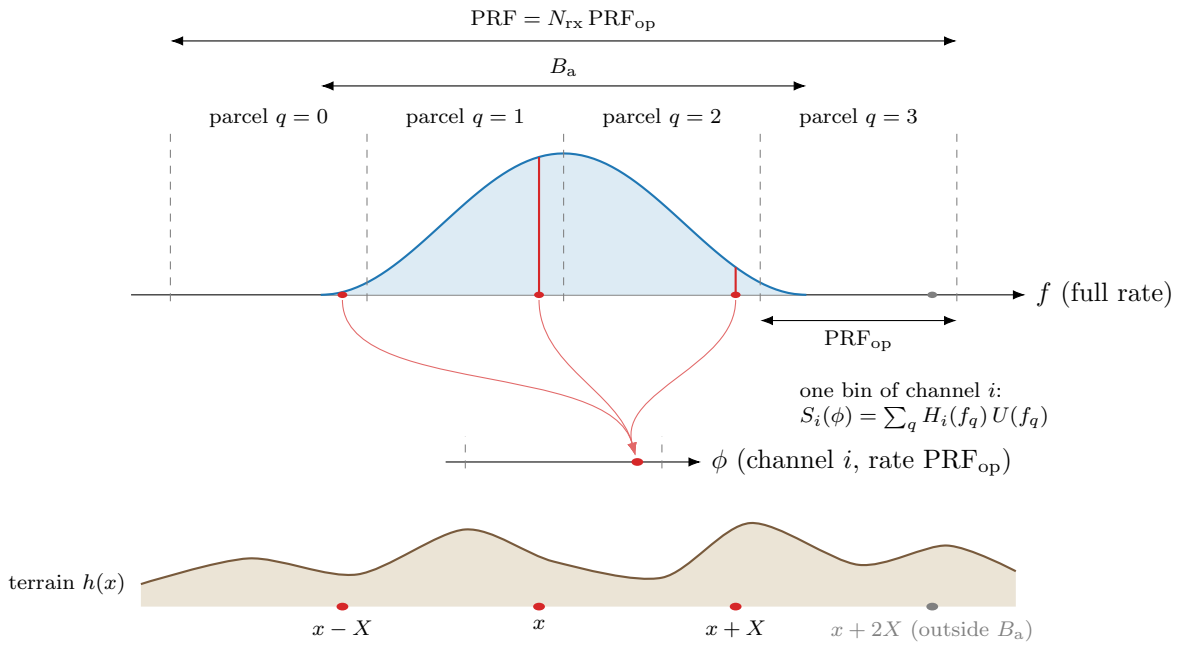


Figure 3.1: Folding of the azimuth spectrum in an aliased channel ($N_{rx} = 4$). The four parcels of one channel bin come, through the ruler (3.3), from ground positions X apart, which in general have different heights.

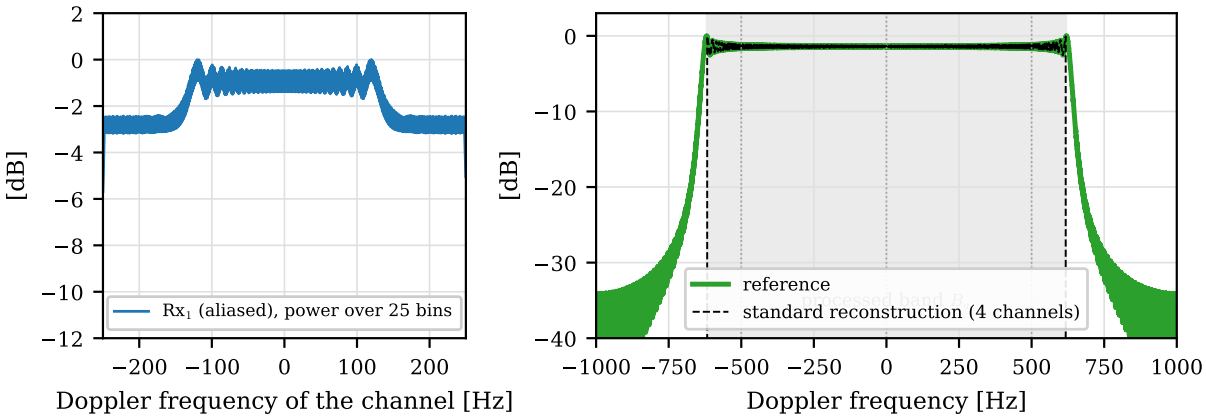


Figure 3.2: One target at $h = 0$, four channels ($b_{xt} = [0, 300, 119.8, 100.5]$ m). Left: power spectrum of one channel at PRF$_{op}$ = 500 Hz (averaged over 25 bins): the folded parcels overlap. Right: spectrum of the reference and of the standard reconstruction at the full PRF; the grey band is the processed band B_a and the dotted lines are the borders of the four parcels, PRF$_{op}$ apart.

c_0 is the gain of the true parcel and c_m, $m \neq 0$, the amplitude that parcel q leaks into parcel $q + m$, which appears in the image as an ambiguity displaced by mX ($0.81\,\mathrm{s}{\cdot}m$). If all channels have the same phase, $c_m = 0$ for $m \neq 0$ and the reconstruction is exact; with cross-track baselines the phases differ and the ambiguity level grows with the spread of $b_{\mathrm{xt},i}h$. Weighted by the fraction of the target bandwidth that falls in the leaking parcels, $|c_m|$ predicts the ambiguity of the uncorrected reconstruction within 0.15 dB for heights above 50 m.

Figure 3.3 shows the effect on one target. At $h = 0$ the channels carry no residual and the standard reconstruction is the reference (up to the band limitation to B_a); at 500 m copies of the target appear at multiples of the ambiguity time, 0.81 s. Figure 3.4 gives the highest ambiguity against the height of the target for three sets of baselines: it grows by 20 dB per decade of height (for small phases $|c_m|$ is proportional to φ_i, hence to $b_{\mathrm{xt}}h$) and saturates between -15 dB and -5 dB once the phases of the channels are of the order of a radian, i.e. above about 100 m with ${b_{\mathrm{xt}}}^{\max} = 300$ m.

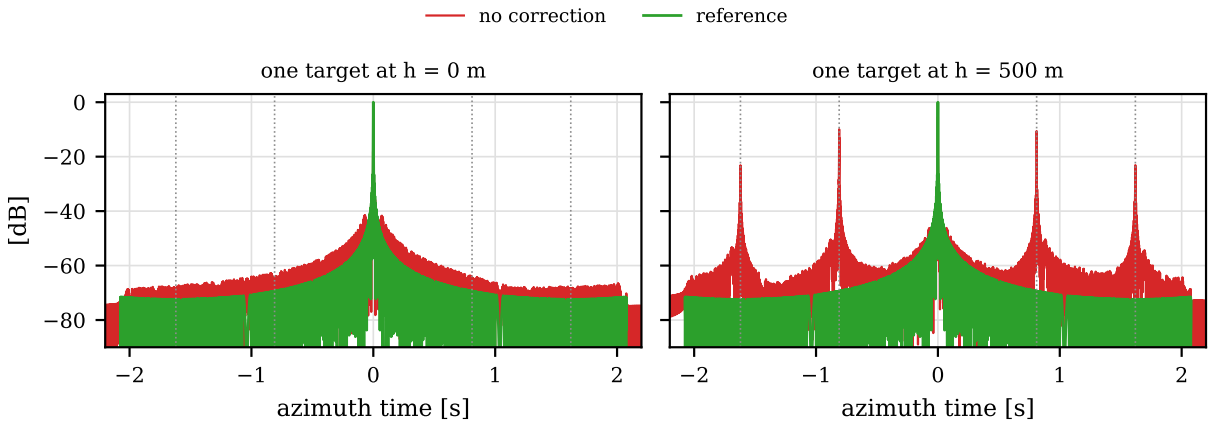


Figure 3.3: Azimuth impulse response of one target ($b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m) after the standard reconstruction (no correction) and of the reference. Dotted lines: the ambiguity positions, ±1 and ±2 times 0.81 s. Left: target at the reference height, the two responses coincide. Right: target at 500 m.

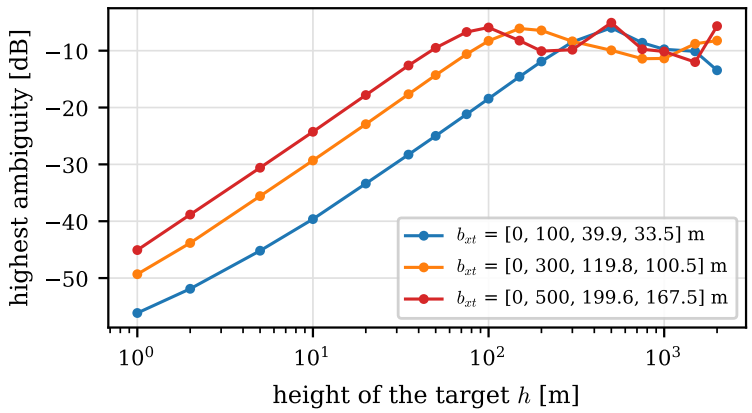


Figure 3.4: Highest ambiguity of the standard reconstruction (no correction) for one target at height h, four channels, three sets of cross-track baselines. At $h = 0$ the three sets give -63.6 dB, the level of the reference limited to the processed band B_a.

The amplitudes c_m are the discrete Fourier transform of the channel phases. This also says what a formation needs for the parcels to be separated once the phases are known: the channels must see them with *different* phases. Cross-track baselines that are not equally spaced, and three or

more channels, give this diversity; it is the property that the joint reconstruction of Chapter 4 relies on.

3.3 Ambiguity metric

All results are reported as the *highest ambiguity*: the largest value of the azimuth impulse response, in dB relative to the peak of the reference, inside ± 20 ms of the ambiguity positions of every target (± 1 and ± 2 times the ambiguity distance of the channel PRF). The reference is the monostatic full-rate signal of the same scene, processed with the same band limitation; its own value (sidelobes of neighbouring targets, band edge) is the floor that any reconstruction can reach. Repeating the measurement with orders up to ± 5 for six and eight channels did not change the conclusions.

4 Joint topography correction and reconstruction

The phase that the terrain adds to each channel, $\varphi_i = 2\pi\Delta C_{0,i}/\lambda$, is known from the DEM and the system geometry (Section 2.4). The joint reconstruction uses it *inside* the inversion: the channels are modelled with the phase of every ground position, and the model and the topography are inverted together. This chapter first shows how the DEM is turned into the phase of every channel, then the two formulations of the method, and finally how the system is built and solved.

4.1 From the DEM to the phase of every channel

The residual $\Delta C_{0,i}$ depends on the height and on the channel, not on the azimuth position (Section 2.4). It is therefore computed once per channel on a table of heights: for 33 heights h_q equally spaced between the lowest and the highest point of the DEM, a virtual point is placed on the iso-range surface of the reference point at that height, and its $\Delta C_{0,i}(h_q)$ is obtained from the exact range histories. A cubic polynomial fitted to the 33 values gives $\Delta C_{0,i}(h)$ for any height, and evaluating it at the DEM height of every position x of the line gives the phase map

$$\varphi_i(x) = \frac{2\pi}{\lambda}\,\Delta C_{0,i}\big(h(x)\big). \tag{4.1}$$

The map is zero for the transmitter channel and, with a flat DEM, constant for every channel. It is the only information about the terrain that the reconstruction receives.

4.2 First formulation: in the sub-apertures

Through the ruler (3.3), the parcel q of a channel bin comes from the ground position $x_q = x_c + \frac{\lambda r_0}{2v} f_q$ (Figure 3.1). The first formulation cuts the line into sub-apertures of 40 ms and, for every sub-aperture and every bin ϕ of the N_{rx} channel spectra, writes the model (3.2) with the phase of each parcel at its own ground position:

$$S_i(\phi) = \sum_q H_i(f_q)\,\exp\!\Big(-\mathrm{j}\frac{2\pi}{\lambda}\Delta C_{0,i}(x_q)\Big)\,U_q. \tag{4.2}$$

The system is solved for the parcels U_q, which are then placed at their true Doppler, transformed back and overlap-added at the full PRF (Figure 4.1). With $\Delta C_0 = 0$ this is the standard reconstruction ($\sim 10^{-7}$), and with one target it equals the oracle. Each parcel now receives the phase of its own position, which is the key step. In this formulation the DEM is read once per bin of a block, i.e. at a ground resolution of a few hundred metres; the second formulation applies it at every position of the line.

4.3 Final formulation: over the whole line

The final formulation removes the sub-apertures. The topographic residual of a scatterer is a constant phase over its whole aperture (Chapter 2); in the *focused* full-rate line $g(x)$ it is

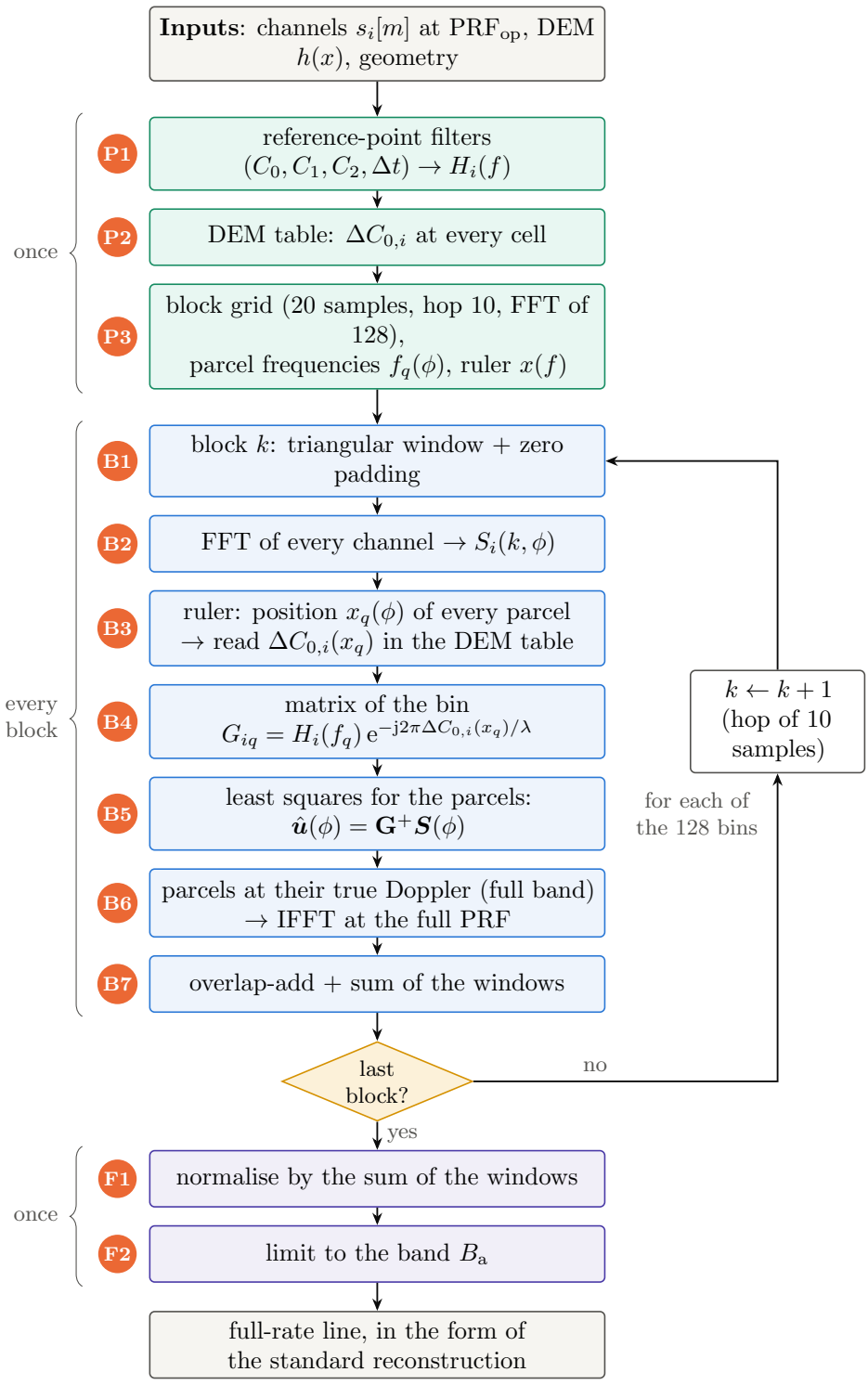


Figure 4.1: The joint reconstruction in the sub-apertures. Green: preparation, done once. Blue: the loop over the blocks of 20 samples; inside each block a small least-squares system is solved for every bin. Violet: finishing. The DEM is read once per bin, at the position the ruler gives to each parcel.

therefore a pointwise multiplication. The channels are modelled as

$$\boldsymbol{s}_i = \mathcal{D}\Big\{ k_i * \big[\, g(x)\, \mathrm{e}^{-\mathrm{j}\varphi_i(x)} \big] \Big\}, \qquad \varphi_i(x) = \frac{2\pi}{\lambda} \Delta C_{0,i}\big(h(x)\big), \tag{4.3}$$

where $\varphi_i(x)$ is the phase map (4.1), from the DEM only, k_i is the response of channel i to the reference point (the azimuth chirp filtered by the reference-point filter H_i of the standard reconstruction) and $\mathcal{D}$ is the decimation to $\mathrm{PRF}_{\mathrm{op}}$ (Figure 4.2). Writing (4.3) as $\boldsymbol{s} = \mathbf{A}\boldsymbol{g}$, the line is estimated by Tikhonov-regularised least squares,

$$\hat{\boldsymbol{g}} = \arg\min_{\boldsymbol{g}} \sum_i \big\| \boldsymbol{s}_i - \mathbf{A}_i \boldsymbol{g} \big\|^2 + \mu^2 \|\boldsymbol{g}\|^2, \tag{4.4}$$

solved with conjugate gradients for least squares (CGLS) [7, 8]. Every iteration costs one application of $\mathbf{A}$ and of its adjoint $\mathbf{A}^{\mathsf{H}}$ (two FFTs per channel each). The regularisation is $\mu = 0.03\,\sigma_{\max}$, with $\sigma_{\max}$ estimated by a few power iterations, and the problem is solved in two passes (iterated Tikhonov: the second pass solves for what the first one left), which turns the filter factor $\sigma^2/(\sigma^2+\mu^2)$ into $1-(1-\sigma^2/(\sigma^2+\mu^2))^2$. Each pass runs up to 60 iterations. The output is re-chirped with the reference response, $U = k * \hat{g}$, so that it has the same form as the standard reconstruction. The complete algorithm is in Figure 4.3.

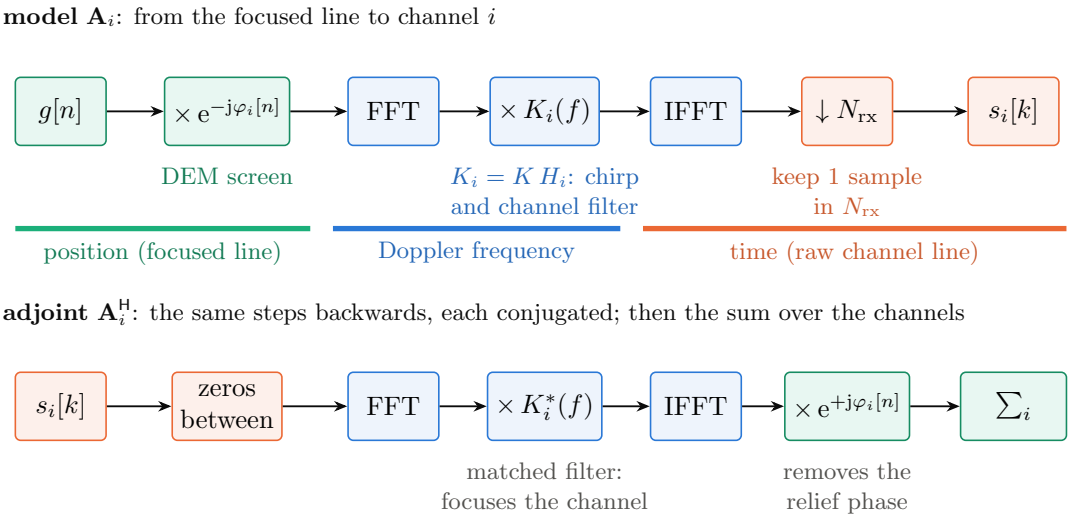


Figure 4.2: The model $\mathbf{A}$ and its adjoint. Each operation is done in the domain where it is simple: the relief phase is a multiplication in *position*, the chirp and the channel filter a multiplication in *Doppler frequency*, the sampling a selection of samples in *time*; the FFTs only move between the domains. The adjoint is the usual SAR processor: it focuses each channel, removes its relief phase and sums the channels.

4.4 Building the model and solving the system

This section builds the model (4.3) from the echo of a single point, one ingredient at a time, writes it as a linear system, shows how that system is solved without ever forming its matrix and explains why the result is the full-rate line $U = K \odot \mathbf{F}\hat{\boldsymbol{g}}$. Unknowns are drawn in red and data in blue.

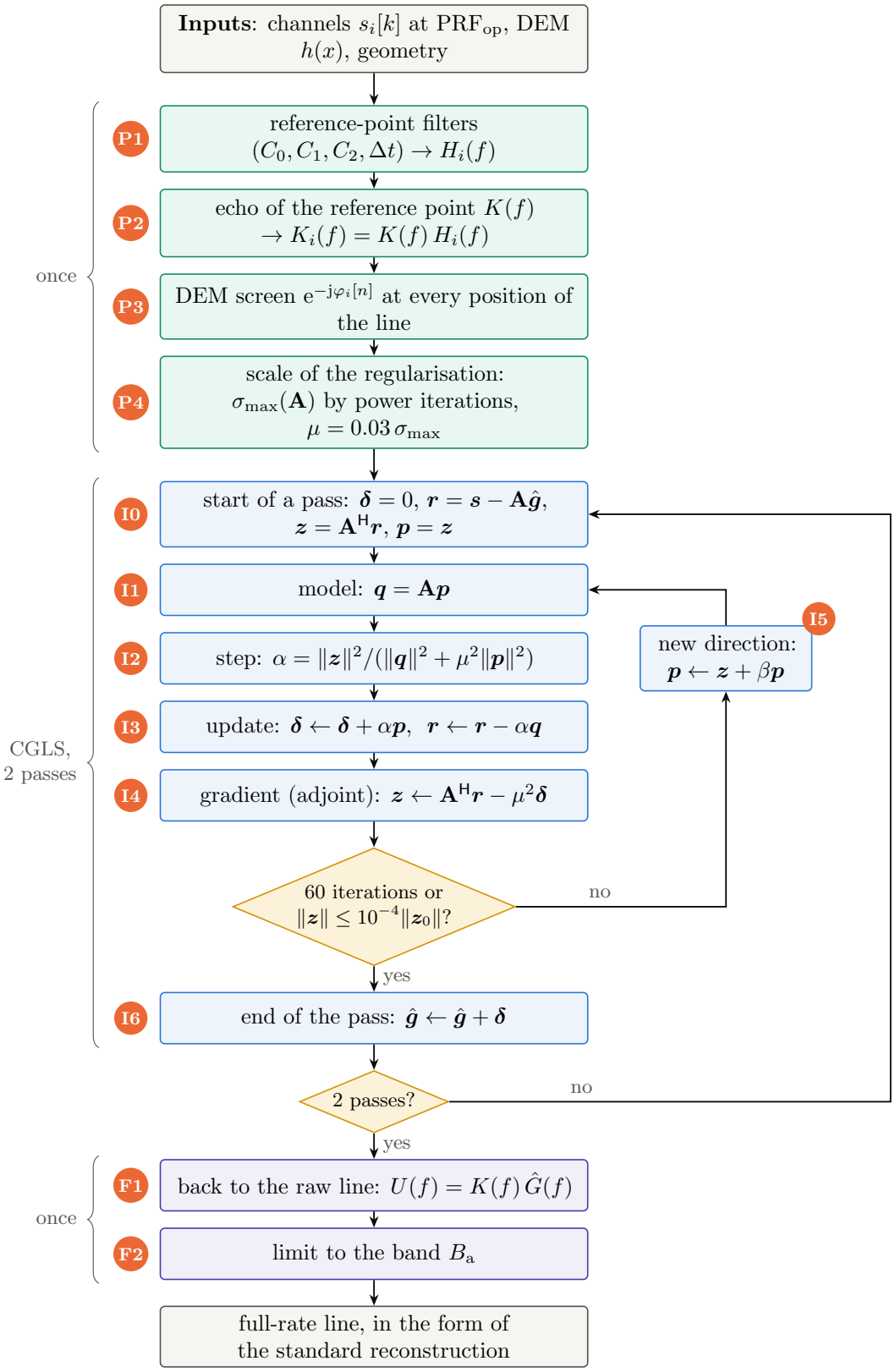
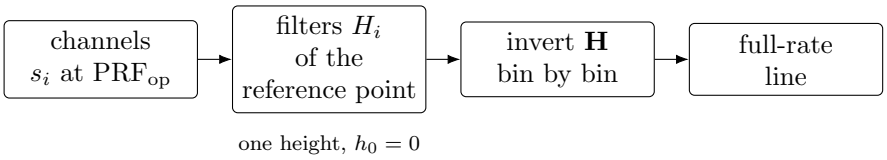


Figure 4.3: The joint reconstruction over the whole line. Green: preparation, done once. Blue: the CGLS iterations, in two passes (iterated Tikhonov); each iteration applies $\mathbf{A}$ and $\mathbf{A}^{\mathsf{H}}$ once (two FFTs per channel each). Violet: back to the raw full-rate line, in the same form as the standard reconstruction.

(a) Standard reconstruction (reference geometry)



(b) Joint reconstruction over the whole line (final version)

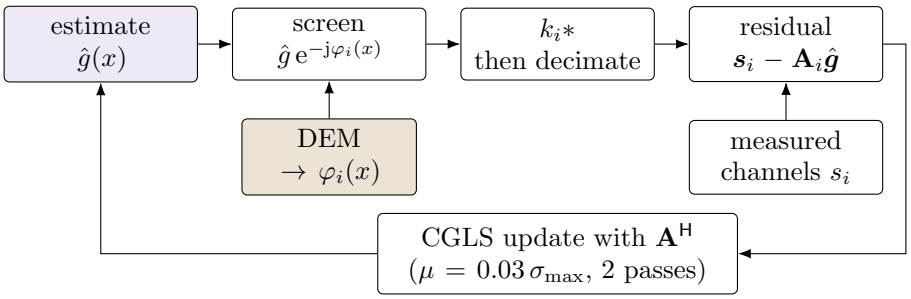


Figure 4.4: Where the DEM phase enters. (a) The standard reconstruction uses the filters of the reference geometry for every scatterer. (b) The joint reconstruction puts the phase of every ground position in the model of the channels and inverts model and topography together.

4.4.1 Building the model

Full-rate line: n is a sample at $\mathrm{PRF} = 2\,000\,\mathrm{Hz}$, $N = 12288$; $\mathcal{F}$ is the DFT; $k[n]$ is the echo of one point (the azimuth chirp) and $K(f) = \mathcal{F}\{k\}$.

Case 1 One target at the reference point, monostatic, sampled at PRF:

$$u[n] = k[n] \xrightarrow{\mathcal{F}} U(f) = K(f).$$

Case 2 Case 1 with the target at $\ell = x/d$, $d = v/\mathrm{PRF} = 3.844\,\mathrm{m}$:

$$u[n] = k[n-\ell] \xrightarrow{\mathcal{F}} U(f) = K(f)\,\mathrm{e}^{-\mathrm{j}2\pi f\ell/\mathrm{PRF}}.$$

Case 3 Case 2 with several targets at ℓ_p, with amplitudes a_p:

$$u[n] = \sum_p a_p\,k[n-\ell_p] \xrightarrow{\mathcal{F}} U(f) = \sum_p a_p K(f)\,\mathrm{e}^{-\mathrm{j}2\pi f\ell_p/\mathrm{PRF}} = K(f)\underbrace{\Big[\sum_p a_p\,\mathrm{e}^{-\mathrm{j}2\pi f\ell_p/\mathrm{PRF}}\Big]}_{G(f)=\mathcal{F}\{g\}},$$

$$g[\ell] = \sum_p a_p\,\delta[\ell-\ell_p]. \tag{4.5}$$

g has one peak per target: it is the focused scene and the unknown of the joint reconstruction.

Case 4 Case 1 with more channels ($i = 0, \ldots, N_{\mathrm{rx}} - 1$): each channel is a filter applied to the monostatic echo,

$$K_i(f) := H_i(f)\,K(f), \qquad H_i(f) = \mathrm{e}^{-\mathrm{j}2\pi\left[C_{0,i}/\lambda + (\Delta t_i - C_{1,i})f + C_{2,i}\lambda f^2\right]} \quad (2.5).$$

Case 5 Case 3 + Case 4:

$$S_i(f) = K_i(f)\, G(f).$$

Case 6 Case 5 with relief: in channel i each target gets the phase $\varphi_{i,p} = \frac{2\pi}{\lambda}\Delta C_{0,i}(\ell_p)$ of (2.7),

$$s_i[n] = \sum_p a_p\, \mathrm{e}^{-\mathrm{j}\varphi_{i,p}}\, k_i[n-\ell_p] \;\xrightarrow{\mathcal{F}}\; S_i(f) = K_i(f) \underbrace{\Big[\sum_p a_p\, \mathrm{e}^{-\mathrm{j}\varphi_{i,p}}\, \mathrm{e}^{-\mathrm{j}2\pi f \ell_p/\mathrm{PRF}}\Big]}_{\mathcal{F}\{\mathrm{e}^{-\mathrm{j}\varphi_i}\, g\}}$$

$$\Longrightarrow \quad \boxed{S_i(f) = H_i(f)\, K(f)\, \mathcal{F}\{\theta_i \odot g\}(f), \qquad \theta_i[\ell] = \mathrm{e}^{-\mathrm{j}\varphi_i[\ell]}}$$

The phase depends on the *position* ℓ (and on the channel), not on the frequency: it is therefore applied to g, as a pointwise *screen* θ_i, *before* the DFT. This is the reason for choosing the focused line as the unknown: in g the relief of every target is one multiplication, at its own position, whereas in the raw line the echo of a target spreads over its whole aperture and mixes with the echoes of the terrain X apart.

Case 7 Case 6 recorded by the channel. Case 6 is the full line (2 000 Hz, $N = 12288$); the channel records one sample in N_{rx} (500 Hz):

$$\boldsymbol{s}_i[k] = s_i^{\mathrm{full}}[N_{\mathrm{rx}}k] \quad\Longrightarrow\quad \boxed{\boldsymbol{s}_i = \underbrace{\mathcal{D}}_{\downarrow N_{\mathrm{rx}}}\, \underbrace{\mathbf{F}^{-1}}_{\mathrm{IFFT}}\, \underbrace{\mathrm{diag}(K_i)}_{\odot K_i}\, \underbrace{\mathbf{F}}_{\mathrm{FFT}}\, \underbrace{\mathrm{diag}(\theta_i)}_{\odot\, \mathrm{e}^{-\mathrm{j}\varphi_i}}\, \boldsymbol{g} = \mathbf{A}_i \boldsymbol{g}} \tag{4.6}$$

with $K_i = H_i \odot K$. Stacking the channels: $\boldsymbol{s} = \mathbf{A}\boldsymbol{g}$, with $\boldsymbol{s} \in \mathbb{C}^{4\times 3072}$ (12288 equations) and $\boldsymbol{g} \in \mathbb{C}^{12288}$ (12288 unknowns). This is (4.3) in matrix form.

4.4.2 What is stored and what is applied

The model and its adjoint are

$$\mathbf{A}_i \boldsymbol{g} = \mathcal{D}\, \mathbf{F}^{-1} \mathrm{diag}(K_i)\, \mathbf{F}\, \mathrm{diag}(\theta_i)\, \boldsymbol{g}, \qquad \mathbf{A}^{\mathsf{H}} \boldsymbol{s} = \sum_i \mathrm{diag}(\theta_i)^{\mathsf{H}}\, \mathbf{F}^{\mathsf{H}}\, \mathrm{diag}(K_i)^{\mathsf{H}}\, (\mathbf{F}^{-1})^{\mathsf{H}}\, \mathcal{D}^{\mathsf{H}}\, \boldsymbol{s}_i, \tag{4.7}$$

with $\mathbf{F}^{\mathsf{H}} = N\mathbf{F}^{-1}$, $(\mathbf{F}^{-1})^{\mathsf{H}} = \mathbf{F}/N$ and $\mathcal{D}^{\mathsf{H}}$ the insertion of $N_{\mathrm{rx}} - 1$ zeros between samples. The adjoint runs the same chain backwards with every factor conjugated, and the channels are summed at the end (Figure 4.2): it focuses each channel with its own filter, removes its relief phase and adds the channels, i.e. it is the usual SAR processor.

$\mathbf{A}$ is a 12288×12288 complex matrix: stored, it would take $12288^2 \times 16\,\mathrm{B} \approx 2.4\,\mathrm{GB}$ and each product would cost about 10^8 operations. Every factor of (4.6), however, is either diagonal or a fast transform, so $\mathbf{A}$ and $\mathbf{A}^{\mathsf{H}}$ are applied as sequences of operations and only the diagonals are kept (Table 4.1): about 1.6 MB and eight FFTs per product with four channels.

Without relief the screens are constant, $\mathbf{F}\,\mathrm{diag}(\theta_i)\,\mathbf{F}^{-1}$ is a multiple of the identity and the only coupling between Doppler bins is the folding of $\mathcal{D}$, which links a bin f only to $f \pm \mathrm{PRF}_{\mathrm{op}}$, $f \pm 2\mathrm{PRF}_{\mathrm{op}}, \ldots$ Ordered by bin, $\mathbf{A}$ splits into 3072 independent 4×4 blocks (one per bin; rows: channels, columns: parcels). This is exactly the standard reconstruction (3.2), and the only place in the project where a matrix is actually assembled and inverted.

Table 4.1: The objects of the model (4.6), and whether each one is ever assembled ($N = 12288$, four channels).

object	meaning	size	assembled?
$\boldsymbol{g}$, $\boldsymbol{s}$	focused scene, channels	N, $4 \times N/N_{\mathrm{rx}}$	vectors
$\mathrm{diag}(\theta_i)$	DEM screen of channel i	$N \times N$, diagonal	only the diagonal
$\mathrm{diag}(K_i)$, $K_i = H_i \odot K$	chirp and channel filter	$N \times N$, diagonal	only the diagonal
$\mathbf{F}$, $\mathbf{F}^{-1}$	DFT	$N \times N$	no: FFT, $\mathcal{O}(N \log N)$
$\mathcal{D}$, $\mathcal{D}^{\mathsf{H}}$	decimation / zero insertion	$N/N_{\mathrm{rx}} \times N$	no: index selection
$\mathbf{A}$, $\mathbf{A}^{\mathsf{H}}$	model and adjoint	$N \times N$	no: chains of the above
$\mathbf{A}^{\mathsf{H}}\mathbf{A} + \mu^2\mathbf{I}$	normal matrix	$N \times N$	never
μ^2	regularisation	scalar	a number

With relief $\mathbf{F}\,\mathrm{diag}(\theta_i)\,\mathbf{F}^{-1}$ is a convolution in Doppler with the spectrum of the screen: it is full and mixes the bins, so the system no longer separates and the whole line is solved at once, iteratively. Figure 4.5 shows one column of $\mathbf{A}^{\mathsf{H}}\mathbf{A}$ for the example of Figure 4.8: on flat terrain a Doppler bin is coupled only with its aliases, more than 40 dB below the bin itself; with the mountain the coupling with the aliases rises to about -22 dB and spreads over ± 7 Hz (some 80 bins).

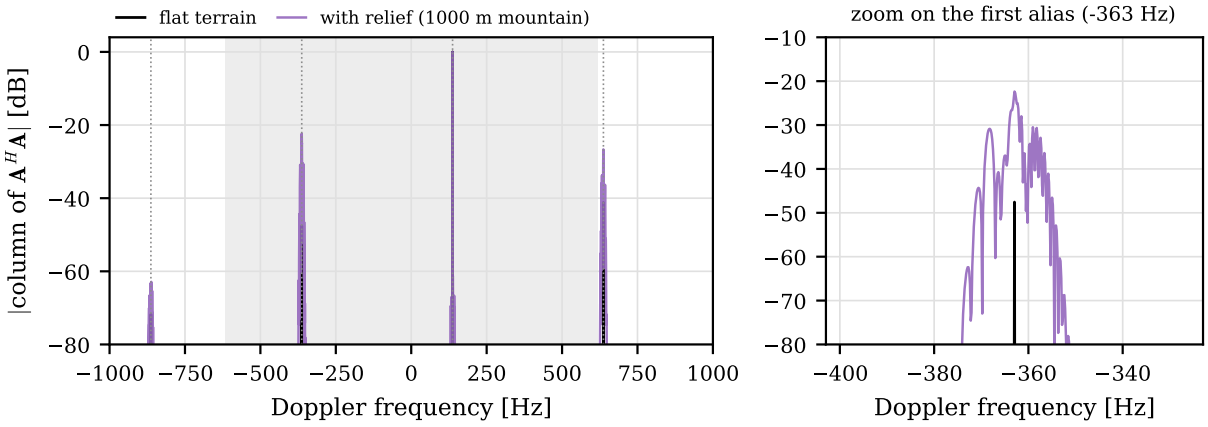


Figure 4.5: One column of $\mathbf{A}^{\mathsf{H}}\mathbf{A}$ in the Doppler domain: the response to a single Doppler bin of g at 137 Hz, for the formation of Figure 4.8, on flat terrain and with the 1 000 m mountain. Dotted lines: the aliases of the bin, $\mathrm{PRF}_{\mathrm{op}} = 500$ Hz apart; grey: the processed band B_{a}. Right: zoom on the first alias.

4.4.3 The problem that is solved

$$\hat{\boldsymbol{g}} = \arg\min_{\boldsymbol{g}} \|\mathbf{A}\boldsymbol{g} - \boldsymbol{s}\|^2 + \mu^2\|\boldsymbol{g}\|^2 \quad \Longleftrightarrow \quad \left(\mathbf{A}^{\mathsf{H}}\mathbf{A} + \mu^2\mathbf{I}\right)\hat{\boldsymbol{g}} = \mathbf{A}^{\mathsf{H}}\boldsymbol{s}, \qquad \mu = 0.03\,\sigma_{\max}(\mathbf{A}). \tag{4.8}$$

The equivalence follows by setting the gradient of the cost to zero, $\mathbf{A}^{\mathsf{H}}(\mathbf{A}\boldsymbol{g} - \boldsymbol{s}) + \mu^2\boldsymbol{g} = \mathbf{0}$. $\sigma_{\max}$ is estimated by the power method, $\boldsymbol{x} \leftarrow \mathbf{A}^{\mathsf{H}}\mathbf{A}\boldsymbol{x}/\|\mathbf{A}^{\mathsf{H}}\mathbf{A}\boldsymbol{x}\|$ (12 iterations), $\sigma_{\max} \approx \sqrt{\|\mathbf{A}^{\mathsf{H}}\mathbf{A}\boldsymbol{x}\|}$, which again uses only $\mathbf{A}$ and $\mathbf{A}^{\mathsf{H}}$.

Why regularise ($\mu^2\|\boldsymbol{g}\|^2$). Writing $\mathbf{A}$ through its singular values, $\mathbf{A}\boldsymbol{v}_k = \sigma_k\boldsymbol{u}_k$, every direction $\boldsymbol{v}_k$ of the scene is recovered with its own weight:

$$\hat{\boldsymbol{g}} = \sum_k \underbrace{\phi_k}_{\text{weight}} (\boldsymbol{u}_k^{\mathsf{H}}\boldsymbol{s})\,\boldsymbol{v}_k, \qquad \phi_k = \begin{cases} 1/\sigma_k & \text{without } \mu \text{ (inverse)}, \\ \dfrac{\sigma_k}{\sigma_k^2 + \mu^2} & \text{with } \mu \text{ (Tikhonov)}. \end{cases} \tag{4.9}$$

A small σ_k is a direction that the channels hardly measure, for example parcels that the channels see with almost the same phases (the conditioning discussed in Section 3.2). Without μ, the factor $1/\sigma_k$ amplifies in those directions the noise and the model errors (the echo shape that the model ignores, the interpolation of the height table, rounding), and $\hat{\boldsymbol{g}}$ explodes. With μ: for $\sigma_k \gg \mu$, $\phi_k \approx 1/\sigma_k$, the same as the inverse; for $\sigma_k \ll \mu$, $\phi_k \approx \sigma_k/\mu^2 \to 0$, the direction is damped. With $\mu = 0.03\,\sigma_{\max}$ only what the channels measure with less than about 3 % of the best direction is damped; the second pass recovers part of what μ removed where the measurement is good.

4.4.4 Conjugate gradients for least squares

$\mathbf{A}^{\mathsf{H}}\mathbf{A} + \mu^2\mathbf{I}$ is Hermitian and positive definite, so (4.8) can be solved by conjugate gradients. CGLS does it with products by $\mathbf{A}$ and $\mathbf{A}^{\mathsf{H}}$ only, never forming the normal matrix (which would also square the condition number in finite precision). The loop is in Figure 4.3:

- each iteration costs one model ($\boldsymbol{q} = \mathbf{A}\boldsymbol{p}$) and one adjoint ($\boldsymbol{z} = \mathbf{A}^{\mathsf{H}}\boldsymbol{r} - \mu^2\boldsymbol{\delta}$); $\alpha = \|\boldsymbol{z}\|^2/(\|\boldsymbol{q}\|^2 + \mu^2\|\boldsymbol{p}\|^2)$ is the optimal step along the direction $\boldsymbol{p}$, and $\boldsymbol{p} \leftarrow \boldsymbol{z} + \beta\boldsymbol{p}$, $\beta = \|\boldsymbol{z}_{\text{new}}\|^2/\|\boldsymbol{z}\|^2$, is conjugate to the previous directions, so no direction is searched twice;
- a pass stops when $\|\boldsymbol{z}\| \le 10^{-4}\|\boldsymbol{z}_0\|$ or after 60 iterations, and adds its result to $\hat{\boldsymbol{g}}$;
- two passes: the second solves the same problem for the residual left by the first and adds its solution to $\hat{\boldsymbol{g}}$ (iterated Tikhonov). The weight becomes $\left[1 - \left(\mu^2/(\sigma_k^2 + \mu^2)\right)^2\right]/\sigma_k$, closer to $1/\sigma_k$ where the channels measure well and still zero where they do not;
- on the example of Figure 4.8: 120 iterations in total, with -40.3 dB of the channels left unexplained.

4.4.5 The answer: $U = K \odot \mathbf{F}\hat{\boldsymbol{g}}$

What is wanted. The line that an ideal *monostatic* radar, at the transmitter, would record at PRF = 2 000 Hz, without aliasing. In the simulation this is the reference. $\hat{\boldsymbol{g}}$ is the scene; U is that scene recorded again by the ideal radar.

Where it comes from. It is Case 3, (4.5), applied to the scene that was found:

$$U(f) = \underbrace{K(f)}_{\text{echo of one point}} \underbrace{\hat{G}(f)}_{\mathcal{F}\{\hat{g}\}} \quad \Longleftrightarrow \quad u[n] = \sum_{\ell} \hat{g}[\ell]\, k[n-\ell], \tag{4.10}$$

i.e. one chirp at each position of the scene, weighted by $\hat{g}$ and summed.

Why without the screen. The screen is the phase of channel i *relative to the monostatic* path; for the monostatic radar itself it is one ($b_{\text{at}} = b_{\text{xt}} = 0 \Rightarrow C_0 = 0 \Rightarrow \Delta C_0 = 0 \Rightarrow \theta \equiv 1$). The effects of the topography are therefore split as in Table 4.2: the monostatic phase of each target is common to all channels and is part of its amplitude a_p, so it is estimated inside $\hat{\boldsymbol{g}}$ and reaches U, exactly as in the reference; the channel-dependent phase lives in the model $\mathbf{A}$ and is used only to explain the channels. U thus contains what the reference contains, the scene with its monostatic phases, and none of the phases that differ between channels, which are the ones that create the ambiguities.

Table 4.2: How each effect of the topography enters the joint model and the answer U.

effect of the topography	type	in the model	in U?
monostatic range phase, $e^{-j4\pi r_p/\lambda}$	constant per target, same in every channel	in a_p (inside $\boldsymbol{g}$)	yes, through $\hat{\boldsymbol{g}}$
extra phase of channel i, $\varphi_i = 2\pi\Delta C_{0,i}/\lambda$	constant per target, differs per channel	screen θ_i (inside $\mathbf{A}$)	no ($\theta \equiv 1$)
echo shape ($K_a \propto 1/r$; C_1, C_2 with h)	not constant	ignored ($\Delta r/r \approx 0.1\,\%$)	—

Why back to the raw line, and not stop at $\hat{\boldsymbol{g}}$.

- *Fair comparison:* the reference, the standard reconstruction and the joint reconstruction all return the raw line of 12288 samples at 2 000 Hz; the focusing that follows is the same for all of them.
- *Band:* outside $|f| \le B_a/2$ the data do not determine $\boldsymbol{g}$; the multiplication by K and the band cut discard that part.
- *SAR chain:* the reconstruction is a pre-processing step (aliased channels in, one clean channel out); the SAR processor expects raw data.

In practice $U = K \odot \mathbf{F}\hat{\boldsymbol{g}}$, then $U(f) = 0$ for $|f| > B_a/2$ ($B_a = 1\,234.8$ Hz), and the line is $\mathbf{F}^{-1}U$, in the same format as the standard reconstruction.

Afterwards (the same for every method). The focusing is image $= \mathbf{F}^{-1}\{K^*U\} = \mathbf{F}^{-1}\{|K|^2\hat{G}\}$, i.e. $\hat{G}$ with a smooth window inside the band: one peak per target (-3 dB width ≈ 5.8 m). The metric of Section 3.3 is then the highest ghost at the folds relative to the peak of the reference. Figure 4.6 summarises the chain.

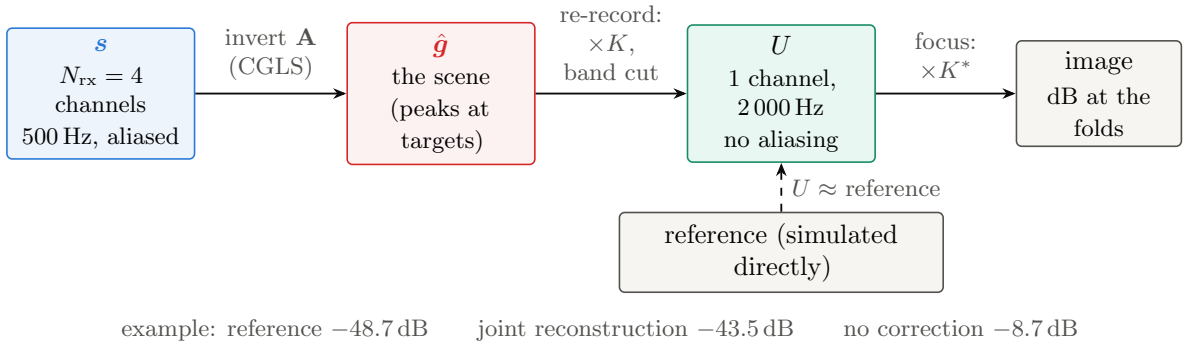


Figure 4.6: From the aliased channels to the image. The joint reconstruction inverts the model to find the scene $\hat{\boldsymbol{g}}$, records it again with the ideal monostatic radar ($\times K$ and band cut) to obtain U, which is compared with the reference simulated directly; the focusing is the same for every method. Values: example of Figure 4.8.

4.5 Properties

- **Consistency.** With a zero DEM the method reproduces the standard reconstruction within 0.1 % rms.

- **Information used.** Only the channels, the system geometry and the DEM. The positions of the targets are never used: removing them from the configuration given to the method does not change a single bit of its output.
- **Self-test of the DEM.** The part of the channels that the model does not explain is $-41\,\mathrm{dB}$ to $-39\,\mathrm{dB}$ with the right DEM and $-20\,\mathrm{dB}$ to $-7\,\mathrm{dB}$ with a wrong one (mirrored or shifted by $1\,\mathrm{km}$), which makes it a natural alarm for DEM errors.
- **Cost.** About $3.3\,\mathrm{s}$ per line with four channels in the 1-D simulator (median over 680 runs): a few tens of FFTs per channel and iteration.
- **Convergence.** Figure 4.7 follows the iterations on the example of Figure 4.8. The first iteration is the adjoint, i.e. the channels focused, corrected and summed ($-10.2\,\mathrm{dB}$); about ten iterations bring the ambiguities below $-39\,\mathrm{dB}$, and the second pass gains another $1\,\mathrm{dB}$ ($-43.5\,\mathrm{dB}$ after 2×60 iterations). The unexplained part of the channels falls to $-38\,\mathrm{dB}$ in the same way.
- **Stability.** The regularisation keeps the solution stable in every configuration: the directions that the channels measure well are recovered, and the few that they hardly measure are damped instead of amplifying noise. Three or more channels with cross-track baselines that are not equally spaced make almost every direction well measured (Section 3.2).

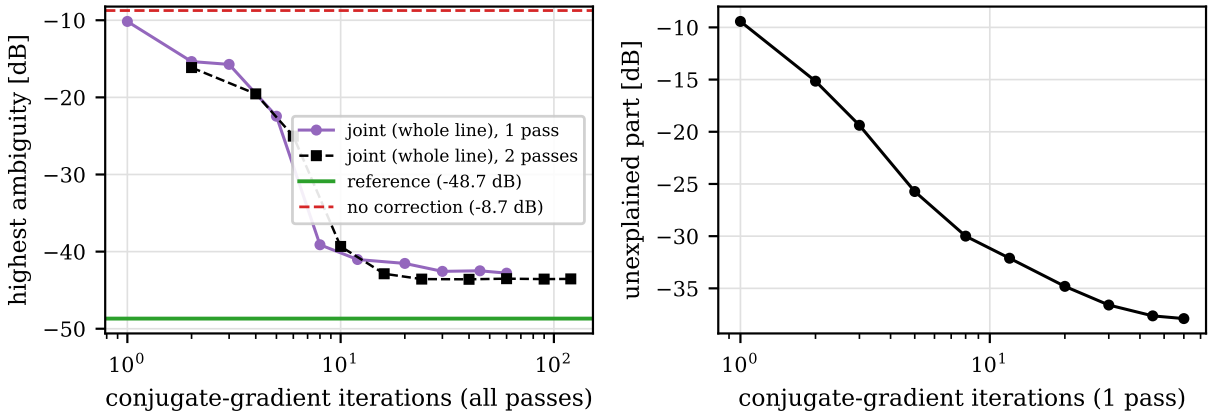


Figure 4.7: Joint reconstruction over the whole line on the example of Figure 4.8 (five targets on a $1\,000\,\mathrm{m}$ mountain, four channels). Left: highest ambiguity against the total number of conjugate-gradient iterations, with one pass and with two passes, against the reference and no correction. Right: part of the channels that the model does not explain, $\|\boldsymbol{s} - \mathbf{A}\hat{\boldsymbol{g}}\|/\|\boldsymbol{s}\|$, during the first pass.

Figure 4.8 shows the example used throughout the development: five targets on a $1\,000\,\mathrm{m}$ Gaussian mountain with $b_{\mathrm{xt}} = [0, 134.9, 20.5, 8.3]\,\mathrm{m}$. The highest ambiguity goes from $-8.7\,\mathrm{dB}$ without correction to $-43.5\,\mathrm{dB}$ with the joint reconstruction; the reference is $-48.7\,\mathrm{dB}$, and $-43.9\,\mathrm{dB}$ once it is limited to the processed band B_{a}, as every reconstruction is.

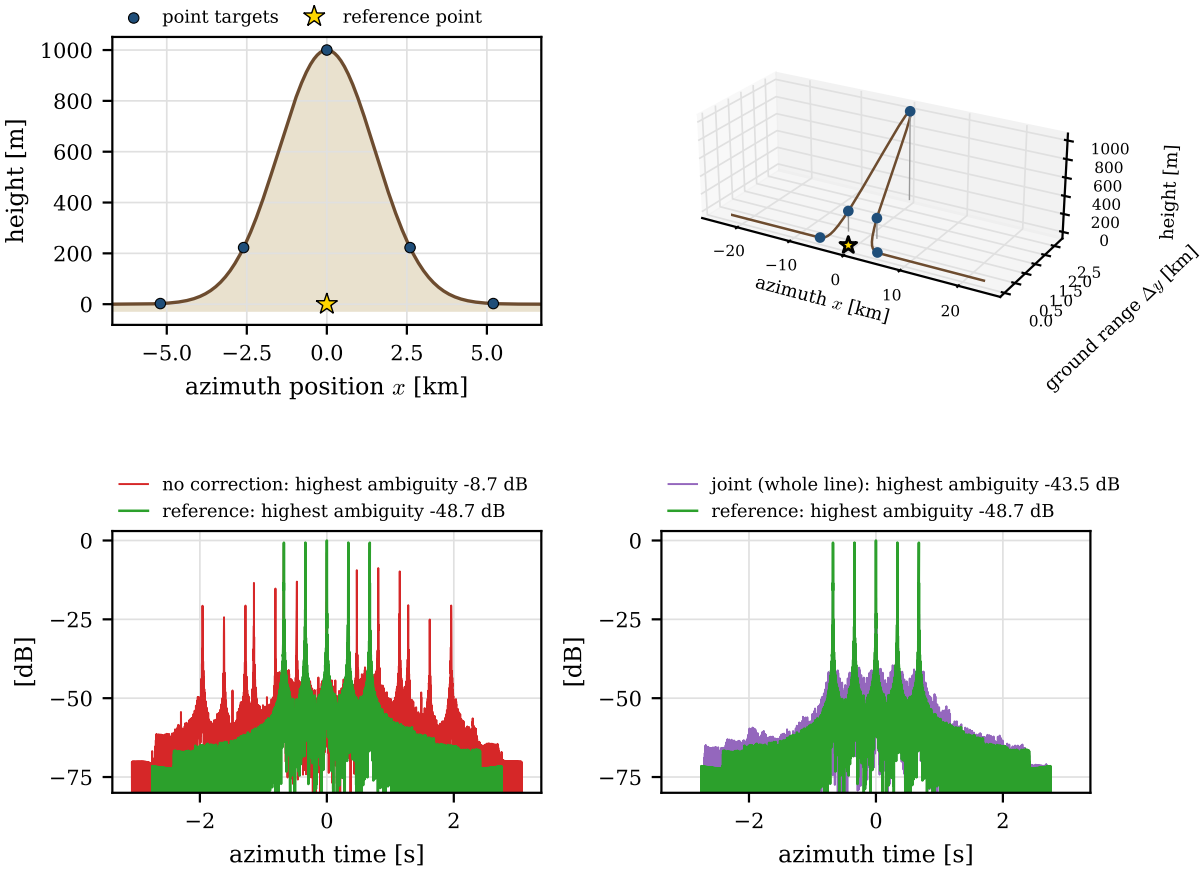


Figure 4.8: Five targets on a 1 000 m Gaussian mountain, four channels, $b_{\mathrm{xt}} = [0, 134.9, 20.5, 8.3]$ m. Top: the targets on the terrain (left) and the whole azimuth line in 3-D (right). Bottom: azimuth impulse responses of the reference with the standard reconstruction (left) and with the joint reconstruction over the whole line (right). The joint reconstruction reads only the DEM.

5 Extension to two dimensions

The 1-D study keeps every target at the slant range of the reference point. In a real scene the terrain varies in range as well, the residual $\Delta C_{0,i}$ depends on both coordinates, the echoes migrate in range and the range envelope of each channel is shifted by half its path difference. A second simulator handles 3-D scenes with along- and cross-track baselines and range-compressed data, and the joint reconstruction is extended to it.

5.1 Model

The unknown is the focused image $g[n, m]$ (Figure 5.2) on azimuth positions x_n (full PRF) and slant ranges of closest approach r_m. The range-compressed data of channel i are modelled as

$$\boldsymbol{s}_i = \mathcal{D}\Big\{ K_i \cdot \mathcal{F}_r\big[\, Q_i \cdot \mathcal{F}_a(S_i\, g)\big]\Big\} \;+\; (\text{envelope term}), \tag{5.1}$$

with

- S_i, the terrain screen at every pixel, $\exp\big(-\mathrm{j}2\pi\Delta C_{0,i}(x, r)/\lambda\big)$, computed from the DEM in radar geometry (the ground range of each pixel is found by a fixed-point iteration, Figure 5.1, which converges for range slopes below $\tan\theta$, i.e. without layover);
- $Q_i(f; r)$, what changes with the range of closest approach in the range-Doppler domain: the azimuth chirp relative to the reference range and the differences of the coefficients $C_0, C_1, C_2, \Delta t$ of the channel;
- $K_i(f, f_r)$, the 2-D kernel: the 2-D FFT of the simulated echo of the reference point in channel i at the full PRF (chirp, range migration, window of the pair, co-registration by $C_{0,i}/2$);
- the envelope term, the first-order shift of the range envelope by $\frac{1}{2}[\Delta C_{0,i} + C_{0,i}(r) - C_{0,i}(r_{\mathrm{ref}})]$, valid while this shift is small compared with the range resolution.

The image is kept on a range grid twice as fine as the data, because a phase that varies with range shifts the range spectrum. The adjoint passes the adjoint test to 2.6×10^{-7}; the solver is the same CGLS as in 1-D ($\mu = 0.03\,\sigma_{\max}$, two passes of up to 60 iterations).

5.2 The common range band

A cross-track baseline shifts the range spectrum seen by the channel, the spectral shift known from interferometry:

$$\nu_i = \frac{c\, b_{\mathrm{xt},i} \cos\theta}{2\lambda\, r \tan(\theta - \alpha)}, \tag{5.2}$$

with α the range slope of the terrain; on flat terrain this is 0.202 MHz per 100 m of baseline. The range frequencies observed by every channel carry all the equations needed to separate the folded parcels, so the output is kept, pixel by pixel, in the range band common to all the channels. In a control case (one target, flat terrain, $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m) the joint reconstruction then reaches -66.1 dB, equal to the reference in the same band. The output is limited to the range frequencies that every channel observes there, and the reference and the uncorrected

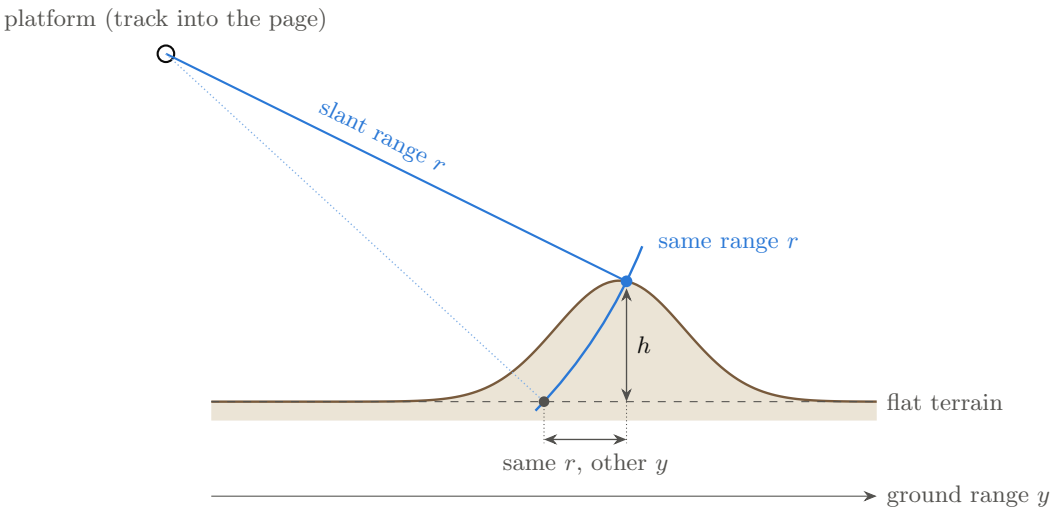
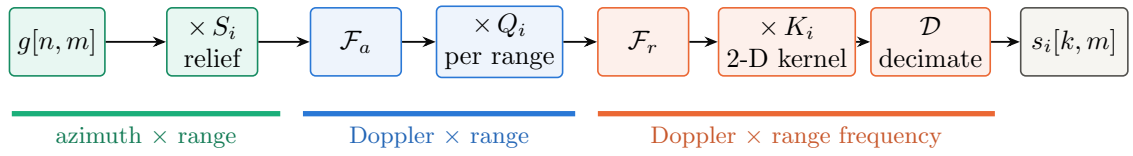


Figure 5.1: The DEM in radar geometry: a pixel at slant range r lies on the terrain at another ground range than on flat terrain. The ground range is found by the fixed-point iteration $y \leftarrow \sqrt{r^2 - [H - h(x,y)]^2}$, which contracts while the range slope stays below $\tan\theta$.

model: from the focused image to the range-compressed data of channel i



adjoint: repeat the spectrum, 2-D matched filter, remove the relief, sum the channels

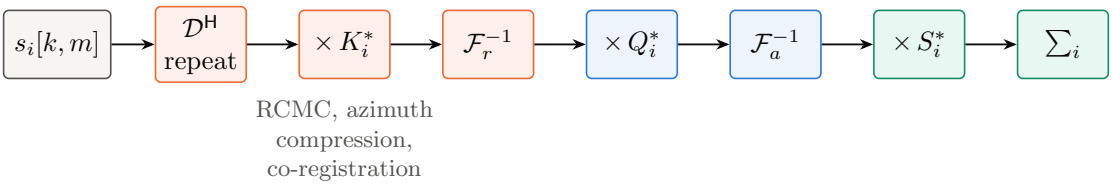


Figure 5.2: The model of the 2-D joint reconstruction and its adjoint (the envelope term is omitted). The relief acts per pixel, the range-dependent focusing per range line in the range-Doppler domain, and the 2-D kernel of the channel in the two-dimensional frequency domain.

reconstruction are compared in the same band. The price is range resolution, a fraction $b_{\mathrm{xt}}/{b_{\mathrm{xt}}}^{\mathrm{crit}}$ of the band; with the 5 MHz bandwidth of the sweeps the median pixel keeps 93 % of the band and the worst pixel 45 %. A wider system bandwidth (e.g. 50 MHz) would reduce this loss about tenfold.

Figure 5.3 shows three 3-D scenes of the 2-D sweep, each with 49 targets spread over 5.4 km of azimuth and 2.4 km of ground range: an oblique ramp and a U-shaped valley with four channels and ${b_{\mathrm{xt}}}^{\mathrm{max}} = 300\,\mathrm{m}$, and the isolated 1 000 m mountain with six channels and ${b_{\mathrm{xt}}}^{\mathrm{max}} = 500\,\mathrm{m}$. Without correction the highest ambiguities are $-10.6\,\mathrm{dB}$, $-12.3\,\mathrm{dB}$ and $-8.6\,\mathrm{dB}$; the joint reconstruction, reading only the DEM, brings them to within 1 dB of the reference in all three. With four channels and $b_{\mathrm{xt}} = [0, 100, 39.9, 33.5]\,\mathrm{m}$ the same mountain also reaches the reference exactly.

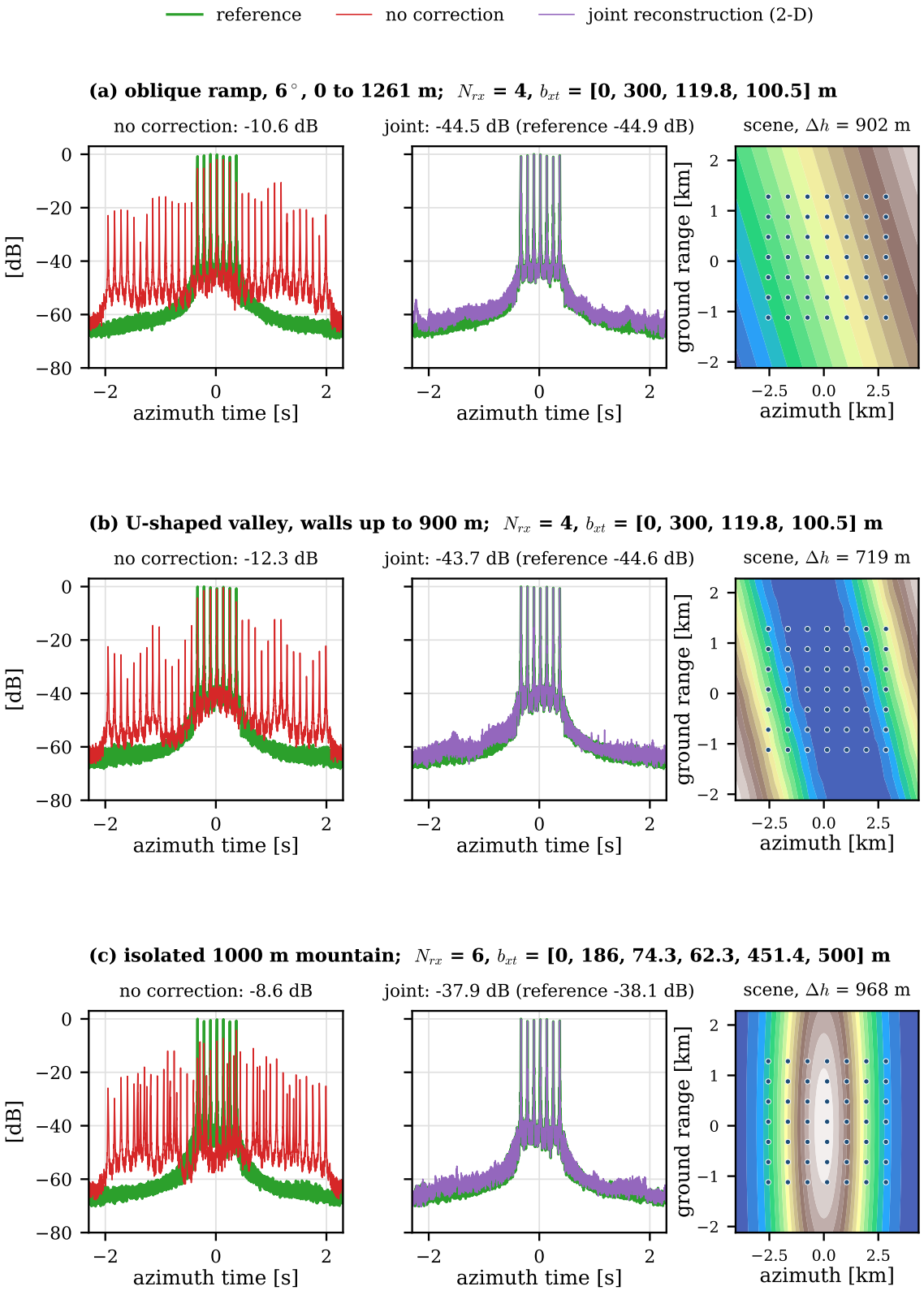


Figure 5.3: Azimuth impulse responses of three scenes of the 2-D sweep (49 targets each, every target focused with its own 2-D matched filter, envelope over the targets; all curves in the same azimuth and common range band). Left: reference and no correction; centre: reference and the 2-D joint reconstruction; right: the terrain seen from above with the targets. The joint reconstruction reads only the DEM. (a) Oblique ramp of 6°, 35° off the flight direction; (b) glacial valley with walls up to 900 m, axis 60° off the flight direction; (c) isolated 1 000 m Gaussian mountain.

6 Results

All numbers in this chapter are the highest ambiguity of Section 3.3, in dB relative to the peak of the reference. The joint reconstruction receives only the system parameters and the DEM. The scenes cover along-track ramps (3° to 20°), sines and cosines of several periods, and realistic terrains (hills, mountains, escarpments, plateaus, fractal alpine relief, glacial valleys, ridges, canyons, volcanoes), each with one, a few and many point targets (up to 49). The baselines are $N_{\mathrm{rx}} = 2, 3, 4, 6, 8$ with ${b_{\mathrm{xt}}}^{\max} = 100\,\mathrm{m}$, 300 m and 500 m (random, the largest exactly ${b_{\mathrm{xt}}}^{\max}$), plus five fixed four-channel sets: two random ones (up to 100 m and up to 500 m), two equally spaced ones ($[0, 10, 20, 30]$ and $[0, 50, 100, 150]$ m) and one with baselines on both sides of the transmitter ($[0, -50, 50, 100]$ m). Channel 0 is always the transmitter.

6.1 Sweep of the joint reconstruction in 1-D

The complete 1-D sweep runs the reference, the uncorrected reconstruction and the joint reconstruction over the whole line on 52 scenes × 20 baseline sets = 1040 runs. The medians are −47.6 dB (reference), −8.4 dB (no correction) and −40.6 dB (joint); 78 % of the runs are within 10 dB of the reference. Figure 6.1 splits the result by the number of channels and Figure 6.2 shows every run.

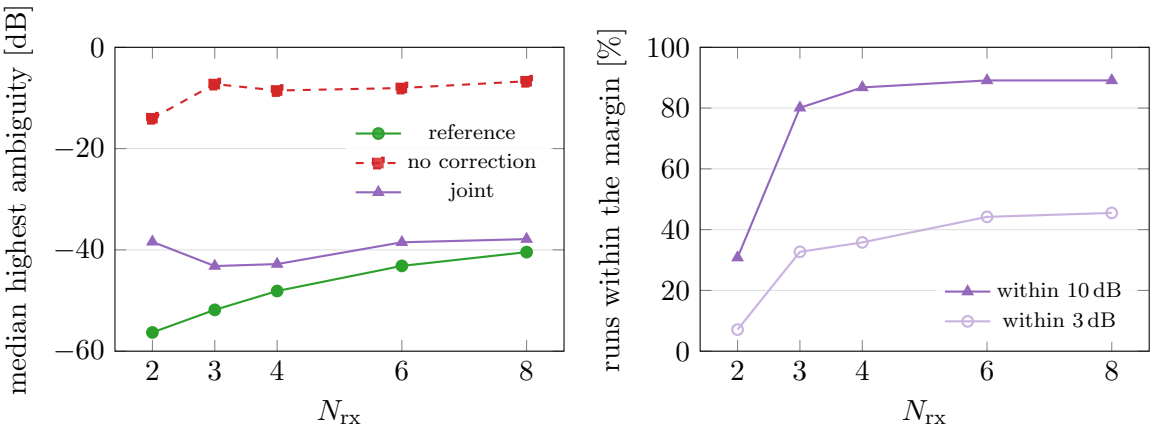


Figure 6.1: 1-D sweep by number of channels. Left: medians of the highest ambiguity. Right: fraction of runs in which the joint reconstruction is within 10 dB and 3 dB of the reference. The reference level rises with N_{rx} because $\mathrm{PRF}_{\mathrm{op}}$ falls and more ambiguity orders fall inside the band.

With random cross-track baselines and three or more channels the joint reconstruction reaches the level of the reference in almost every run: of the 83 runs that stay above −30 dB and more than 10 dB from the reference, only 5 belong to these formations, while the others have two channels (53) or the equally spaced baselines $[0, 50, 100, 150]$ m (25), which give the channels too little phase diversity to separate the parcels (Section 3.2). Figure 6.3 shows the role of the number of channels on one scene with 49 targets on random mountains and ${b_{\mathrm{xt}}}^{\max} = 300\,\mathrm{m}$: with three, four and eight channels the joint reconstruction is within 1 dB to 4 dB of the reference, and with two channels it still lowers the ambiguities to −22.3 dB (reference −38.5 dB).

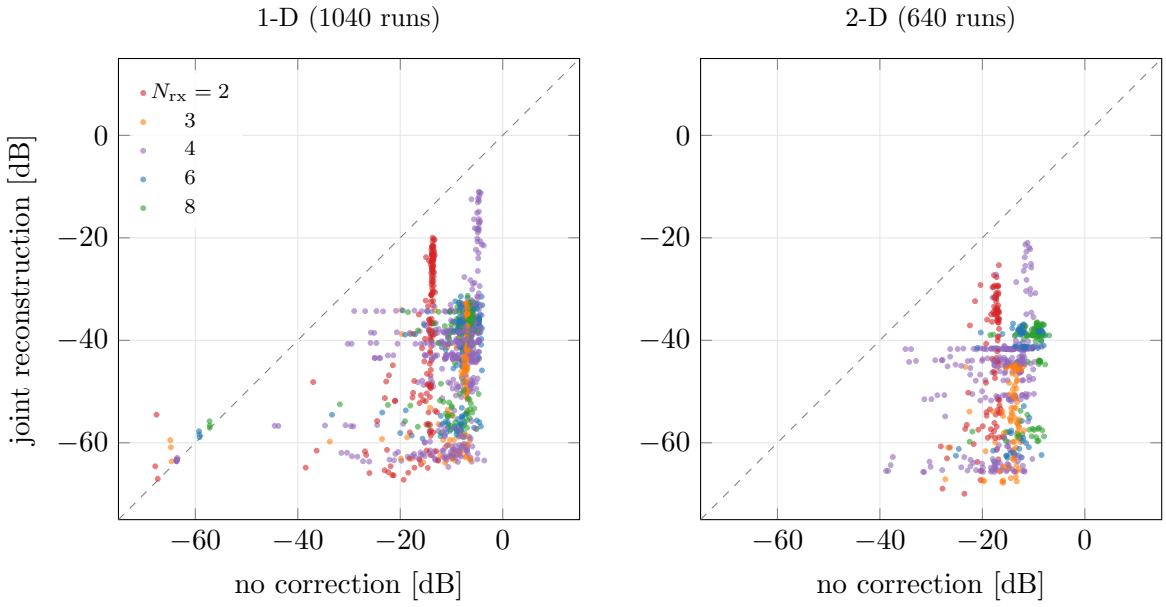


Figure 6.2: Every run of the 1-D and 2-D sweeps: highest ambiguity of the joint reconstruction against the uncorrected reconstruction, coloured by the number of channels. Points below the diagonal are improvements.

6.2 Sweep of the joint reconstruction in 2-D

The 2-D sweep uses 32 3-D scenes (ramps along azimuth, range and oblique; sines and cosines; realistic and fractal terrains without layover; one target on a flank, five on a diagonal or a 7×7 grid) with the same 20 baseline sets, 640 runs in total. The medians are -47.6 dB (reference), -13.9 dB (no correction) and -44.4 dB (joint), all in the common range band. In every run the joint reconstruction is at least as good as the uncorrected one.

Table 6.1: 2-D sweep by number of channels: median gap between the joint reconstruction and the reference, and fraction of runs within 3 dB of the reference.

N_{rx}	runs	reference [dB]	no corr. [dB]	joint [dB]	joint − ref. [dB]	within 3 dB [%]
2	96	−60.8	−17.4	−39.3	22.9	6
3	96	−55.5	−13.8	−53.9	1.1	65
4	256	−45.2	−15.1	−44.6	0.3	84
6	96	−41.5	−12.2	−41.4	0.2	91
8	96	−39.1	−9.5	−39.1	0.2	96

With three or more channels the joint reconstruction is within a fraction of a dB of the reference in the median (Table 6.1), as in 1-D.

6.3 Accuracy of the DEM and noise

DEM accuracy. The phase error is $\approx 9.0 \times 10^{-5}\, b_{xt}\, \delta h$ rad, so the DEM accuracy needed grows with the baseline (Figure 6.4). With an rms height error of 2 m the joint reconstruction keeps -43.6 dB with ${b_{xt}}^{max} = 135$ m, and with 5 m it reaches -36 dB; for ${b_{xt}}^{max} = 300$ m a DEM of about 2 m keeps it near -29 dB.

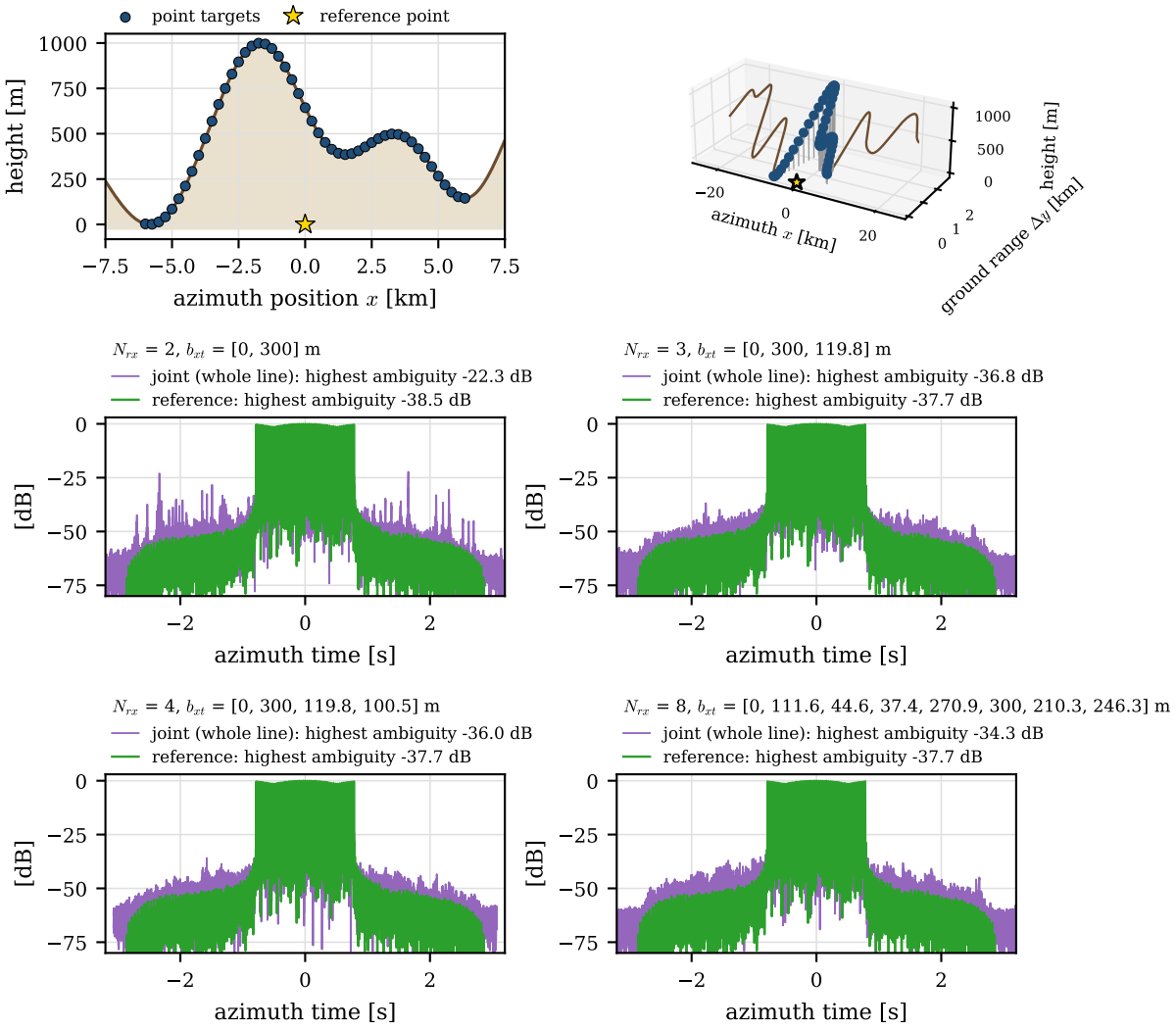


Figure 6.3: 49 targets 250 m apart on random mountains (relief 1 000 m), largest cross-track baseline 300 m, channel 0 on the transmitter. Top: the targets and the terrain near them (left) and along the whole azimuth line (right). Below: impulse responses of the reference and of the joint reconstruction over the whole line for $N_{\mathrm{rx}} = 2$, 3, 4 and 8.

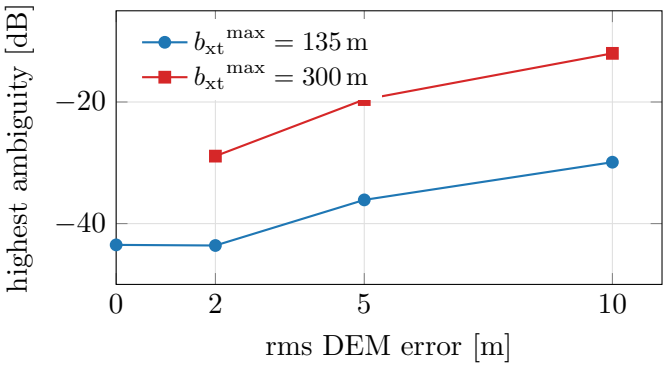


Figure 6.4: Joint reconstruction over the whole line with a random DEM error (five targets on the mountain of Figure 4.8).

To do: Confirm the scene and the error model behind Figure 6.4 (taken from the audit notes) and add the error-free point for $b_{\mathrm{xt}}{}^{\max} = 300\,\mathrm{m}$.

Noise. Relative to the standard reconstruction, the joint reconstruction changes the output noise by 0.3 dB to 4 dB with three or more channels and random baselines; the regularisation keeps it bounded in every configuration.

Formation design. The channels separate the parcels best when they see them with different phases: three or more channels and cross-track baselines that are not equally spaced (Section 3.2). This is a simple rule for the design of a formation that uses the joint reconstruction.

6.4 Real terrain: SRTM lines over the Sierra Nevada

The scenes of the previous sections are synthetic. To check the joint reconstruction over the whole line on real relief, azimuth lines were cut from the SRTM 3″ DEM (about 90 m, WGS 84) [9] over the Sierra Nevada (California), using the file `raster/srtm90.tif` of `github.com/opengeos/data`. Every result is still one azimuth line (1-D): a straight 47 km profile of the DEM, sampled every 90 m and interpolated linearly between samples. The heights are taken relative to the mean height of the line, which is the reference height of the processing. Point targets of unit amplitude are placed on the terrain, the four channels are simulated with $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m (channel 0 on the transmitter), and the joint reconstruction receives the same SRTM profile as its DEM, never the targets. The metric is the highest ambiguity of Section 3.3; the reference is the monostatic full-rate line of the same scene. The script is `sata1d_code/joint_cases/real_dem/run_real_dem.py` and the figures are made by `real_dem_figures.py` in the same folder.

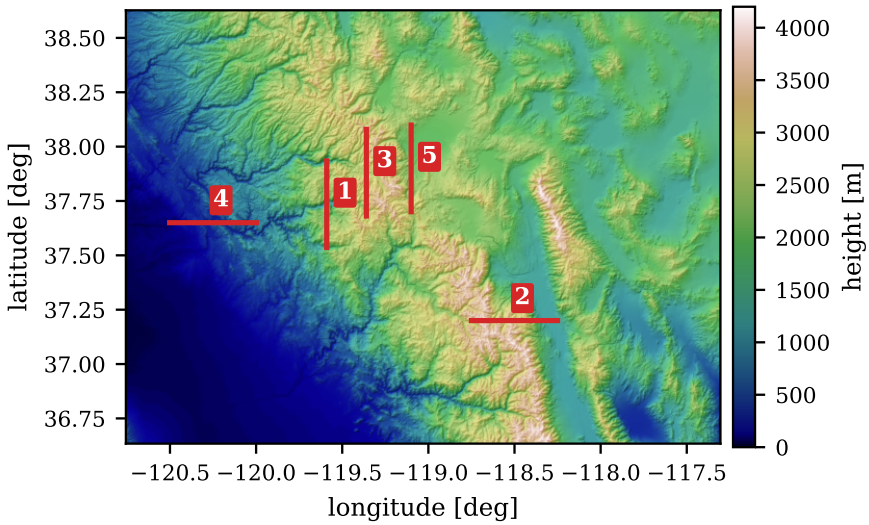


Figure 6.5: SRTM 3″ DEM of the Sierra Nevada with the five named lines of Table 6.2 (red, 47 km each). The map only shows where the lines come from: every result is computed on one line.

Figures 6.6 to 6.8 show three of the lines. Without correction every target leaves ghosts at $\pm X$ and $\pm 2X$ ($X = 6.23$ km) between -8.2 dB and -5.7 dB; the joint reconstruction brings all of them to the level of the reference, within 0.6 dB on the five lines (Table 6.2). The Yosemite line has steep valley walls (about 1 700 m of relief over a few kilometres), the escarpment line goes from the Sierra crest down to the floor of Owens Valley (2 800 m), and the High Sierra line has many short slopes, so that the height, and with it the phase of every channel, changes every few

Table 6.2: The five named SRTM lines with five targets each (at $x = -5.0, -2.4, 0, 2.6, 5.3$ km, chosen so that no ghost at $\pm X$ or $\pm 2X$ falls on another target). Highest ambiguity in dB; the reference is −41.6 dB on every line (sidelobes of the neighbouring targets inside the ghost windows).

Line	Direction	Terrain height [m]	No correction [dB]	Joint [dB]
1 Yosemite Valley	north–south	1213–2893	−5.7	−41.5
2 Eastern Sierra escarpment	west–east	1198–4053	−8.2	−41.5
3 High Sierra, Tuolumne	north–south	2351–3635	−6.0	−41.6
4 Western foothills	west–east	52–932	−6.1	−41.4
5 Mono basin	north–south	1948–3432	−6.2	−41.0

hundred metres. None of these is a difficulty for the joint reconstruction, because it applies the DEM phase at every position of the line.

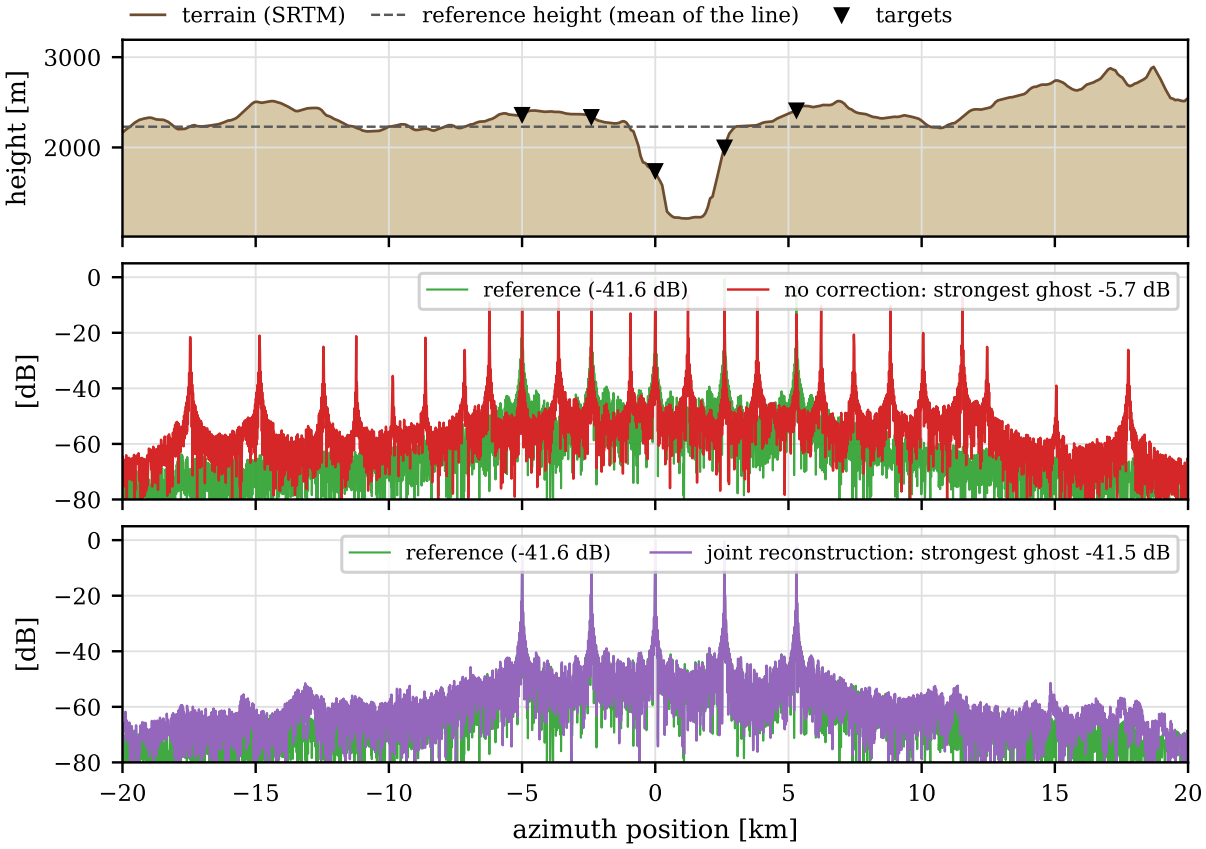


Figure 6.6: Yosemite Valley (north–south, line 1). Top: SRTM profile, reference height and the five targets. Middle and bottom: focused line in dB relative to the peak of the reference, without correction and with the joint reconstruction; the reference (green) is drawn below both.

Figure 6.9 zooms on the Yosemite target that is farthest from the reference height (on the valley floor, 487 m below it). Without correction the peak is 4.4 dB low and 29.7° off, because part of its energy went to the ghosts; the ghost at $+X$ is at −10.2 dB. With the joint reconstruction the peak equals the reference within 0.01 dB and 0.02°, and the ghost drops to −43.1 dB, below the sidelobes of the reference around it.

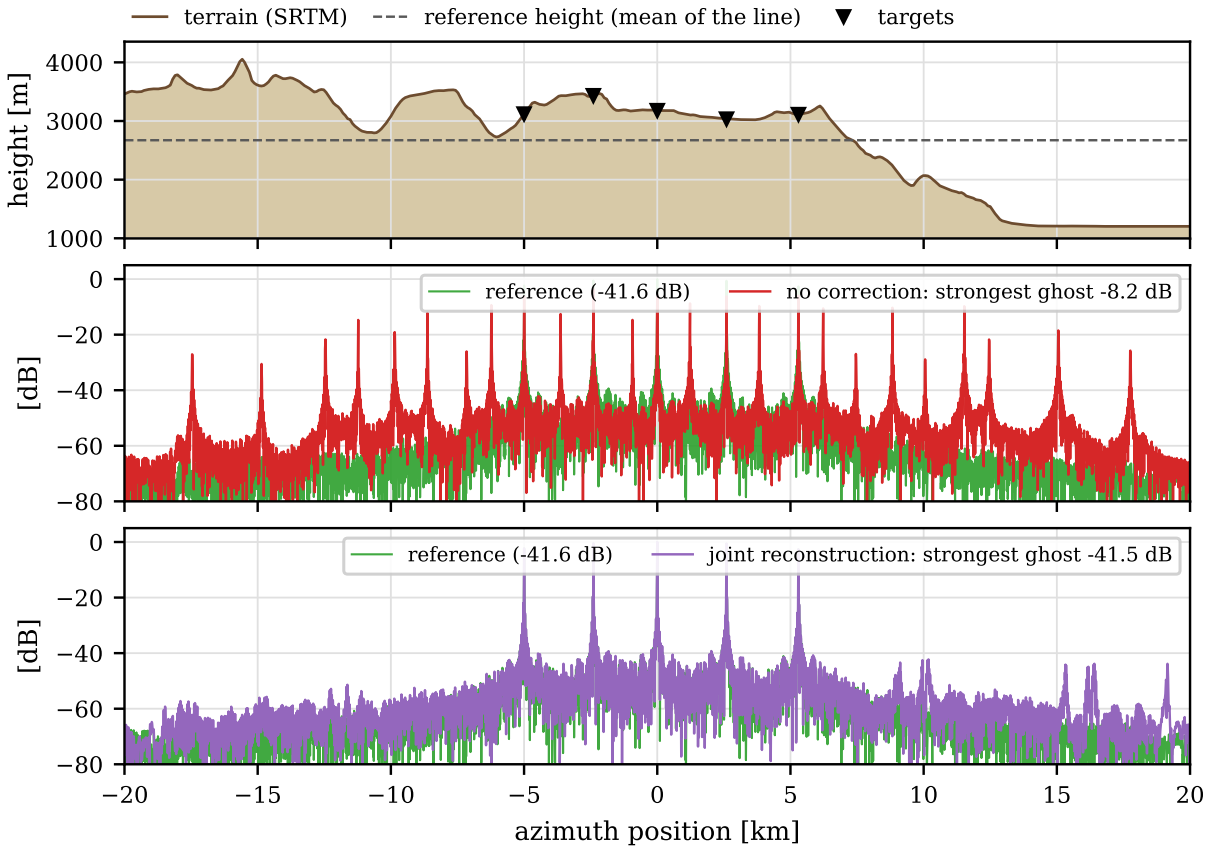


Figure 6.7: Eastern Sierra escarpment (west–east, line 2), from the crest to the floor of Owens Valley; same layout as Figure 6.6.

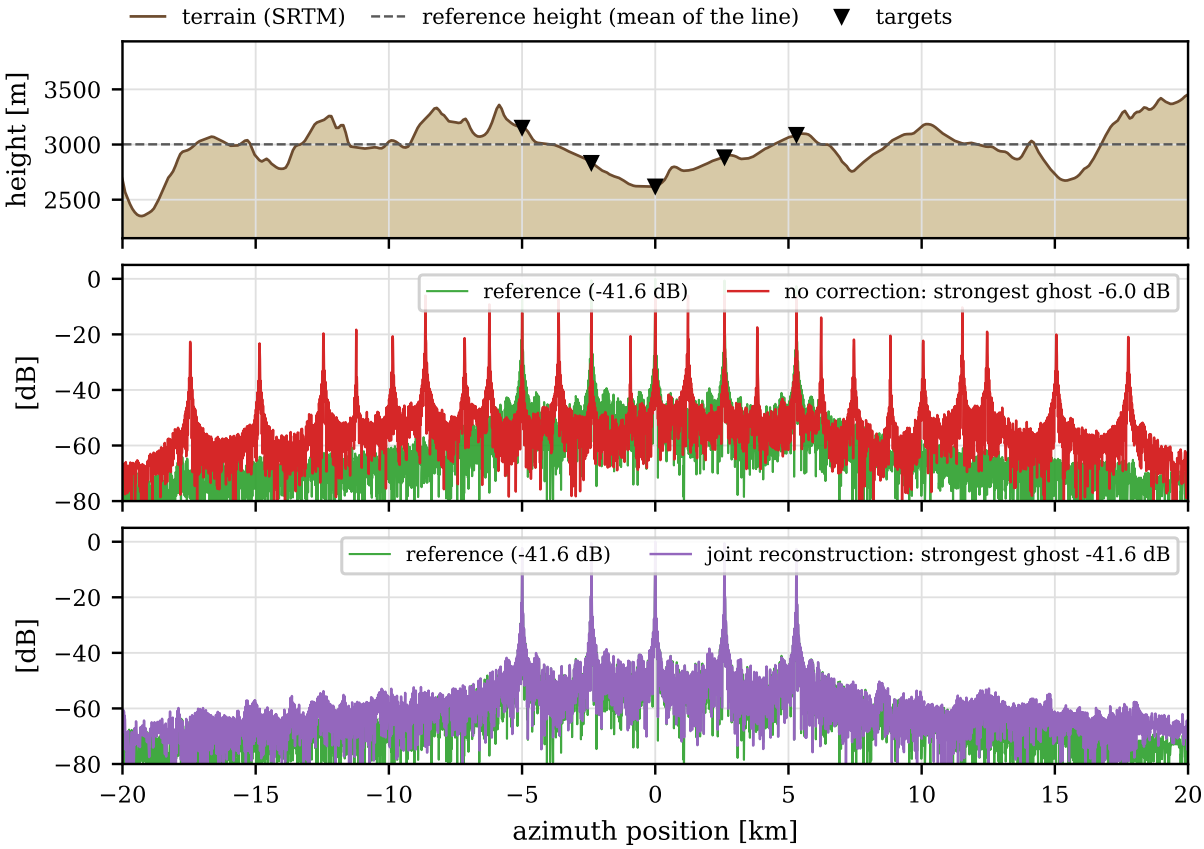


Figure 6.8: High Sierra near Tuolumne (north–south, line 3): rugged peaks and valleys; same layout as Figure 6.6.

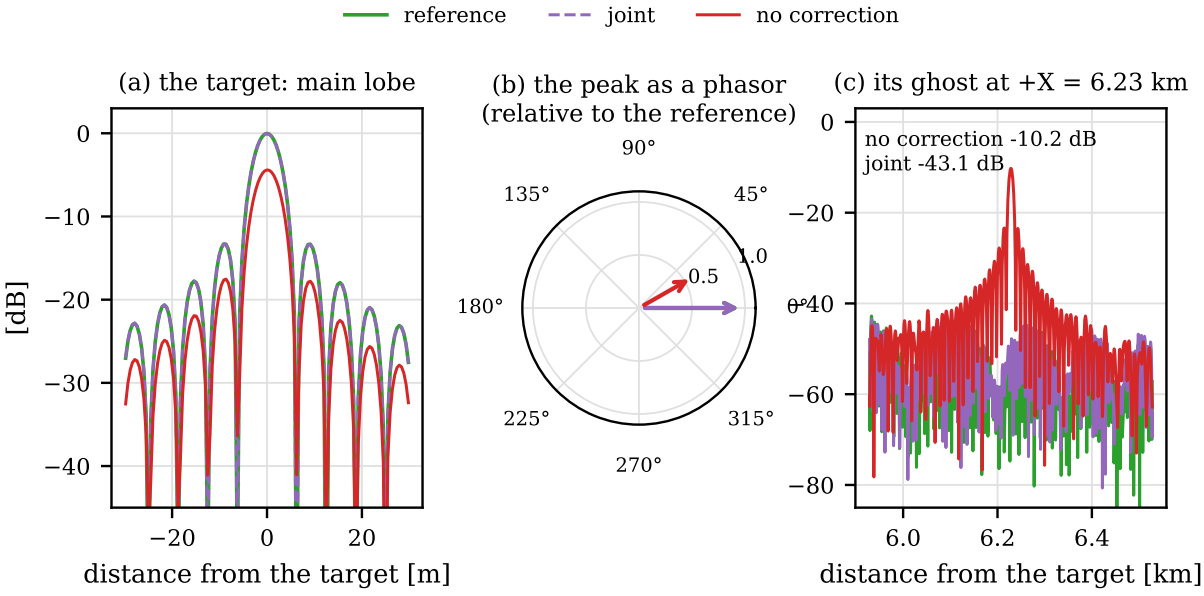


Figure 6.9: Yosemite, valley-floor target 487 m below the reference height. (a) Main lobe; the joint line lies on the reference. (b) Complex peak relative to the reference. (c) The window of its ghost at $+X$. All levels in the processed band B_a, relative to the peak of the reference.

Many targets. With 20 targets every 750 m (from −7.1 km to 7.1 km) every ghost falls between two targets, and the highest ambiguity is the worst of 80 ghost windows per line. The reference reaches only −32.0 dB here, from the sidelobes of the neighbours inside the windows. Without correction the five lines give −5.7 dB to −5.1 dB; the joint reconstruction gives −31.5 dB to −28.1 dB, within 4 dB of the reference (Figure 6.10).

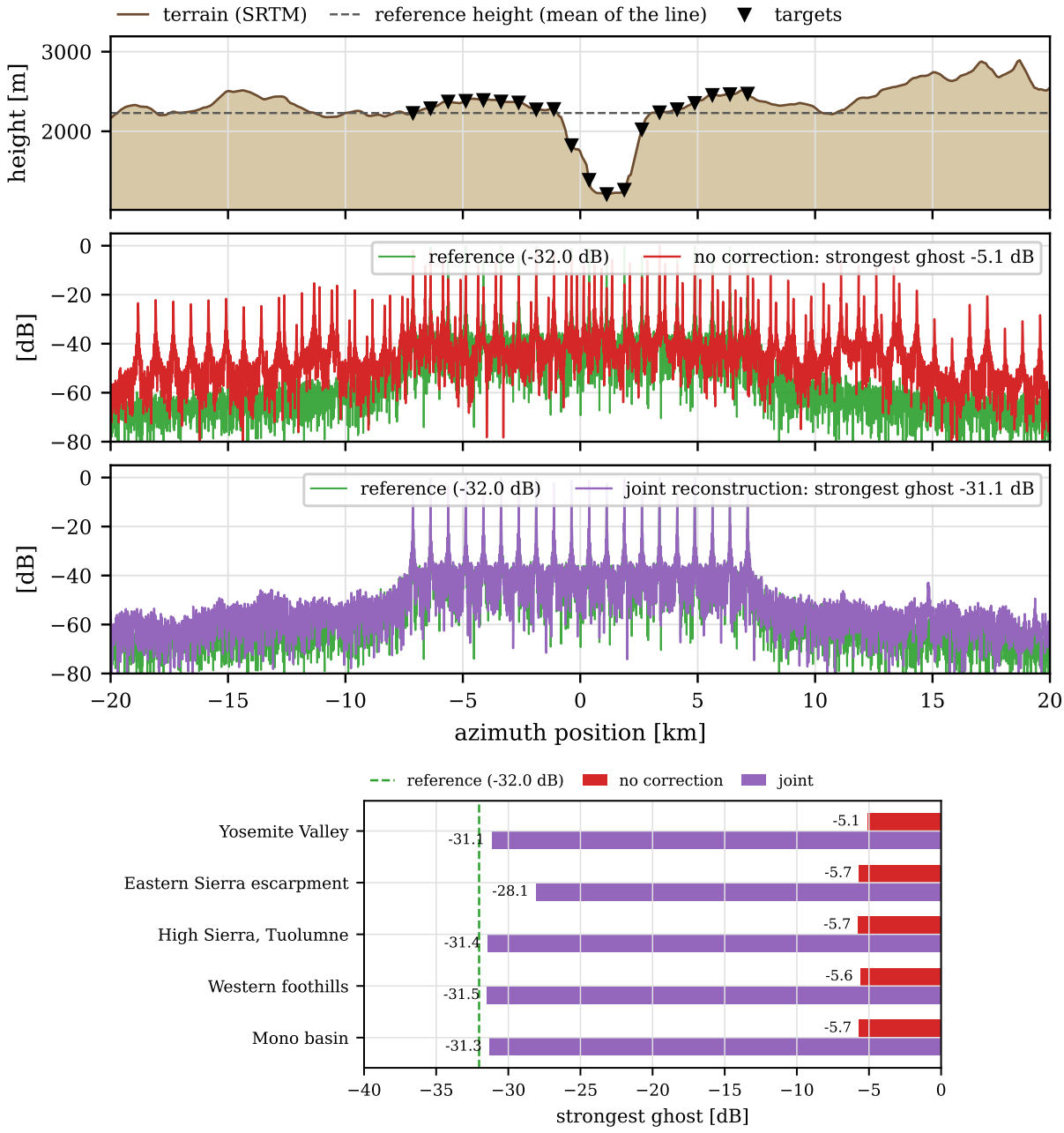


Figure 6.10: Top: Yosemite line with 20 targets every 750 m; same layout as Figure 6.6. Bottom: highest ambiguity of the five lines with 20 targets each.

200 random lines. To avoid choosing the lines by hand, 200 lines were drawn at random over the whole DEM (random centre and direction, the full 47 km inside the raster), with the five targets of Table 6.2. The relief of a line reaches 2 930 m; half of the lines use $b_{\mathrm{xt}} = [0, 300, 119.8, 100.5]$ m and half $b_{\mathrm{xt}} = [0, 134.9, 20.5, 8.3]$ m (largest 135 m). Figure 6.11 gives the result: the median highest ambiguity goes from −8.3 dB without correction to −41.5 dB with the joint reconstruction (reference −41.6 dB), a median gain of 33 dB. In the 192 lines with more than 200 m of relief

the gain is at least 28 dB in 92 % of them, and the joint reconstruction is within 10 dB of the reference in 199 of the 200 lines; on the flattest line (16 m of relief) both methods are already at the reference. The gain grows with the relief, because the uncorrected ghosts grow with it while the joint result stays at the reference.

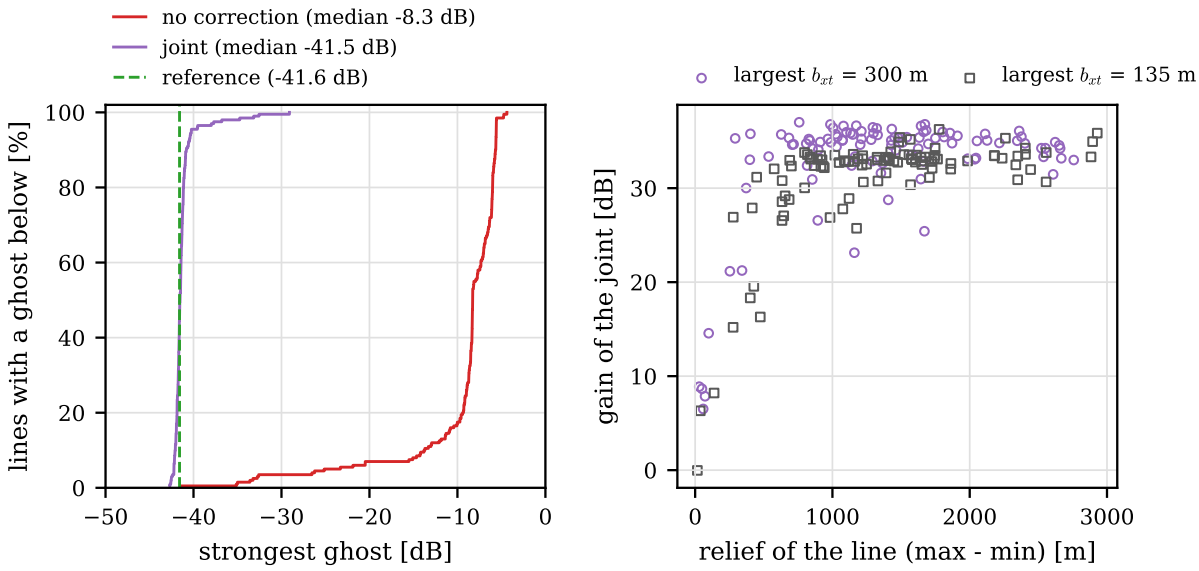


Figure 6.11: 200 random SRTM lines over the Sierra Nevada, five targets each. Left: cumulative distribution of the highest ambiguity. Right: gain of the joint reconstruction over no correction against the relief of the line, for the two baseline sets.

In these runs the DEM given to the reconstruction is the terrain of the simulation; the accuracy needed for a real DEM is discussed in Section 6.3.

7 Conclusions and outlook

7.1 Conclusions

- Cross-track baselines turn topography into a channel-dependent phase (2.7), proportional to $b_{\mathrm{xt},i}h$, which the reconstruction filters of the reference geometry do not include. Without correction the median highest ambiguity of the sweeps is $-8.4\,\mathrm{dB}$ in 1-D and $-13.9\,\mathrm{dB}$ in 2-D, against a reference near $-48\,\mathrm{dB}$.
- This phase is known from the DEM and the system geometry. The joint reconstruction places it inside the model of the channels, at every position of the focused line, and inverts model and topography together by regularised least squares with conjugate gradients. The model and its adjoint are applied with FFTs only, and the method reduces to the standard reconstruction on flat terrain.
- Using only the system parameters and the DEM, the joint reconstruction reaches a median of $-40.6\,\mathrm{dB}$ in 1-D (78 % of the runs within 10 dB of the reference, about 90 % for four or more channels) and $-44.4\,\mathrm{dB}$ in 2-D, where with three or more channels it is within 1 dB of the reference in the median.
- On real terrain (200 azimuth lines cut from the SRTM DEM of the Sierra Nevada, relief up to 2 930 m) the joint reconstruction lowers the median highest ambiguity from $-8.3\,\mathrm{dB}$ to $-41.5\,\mathrm{dB}$, at the level of the reference ($-41.6\,\mathrm{dB}$), and is within 10 dB of it in 199 of the 200 lines (Section 6.4).
- The method works best with three or more channels and cross-track baselines that are not equally spaced, so that the channels see the folded parts of the spectrum with different phases. The accuracy of the DEM sets the final level: the phase error grows as $b_{\mathrm{xt}}\,\delta h$, so long baselines call for accurate DEMs.

7.2 Outlook

- Validation with realistic antenna patterns, orbits and baselines that vary along the orbit, and eventually with real multistatic data.
- 2-D: distributed scenes, layover and shadow, finer range resolution (envelope interpolation) and wide swaths (range-dependent kernels in blocks).
- Joint estimation of a DEM correction, using the unexplained part of the channels as the cost.
- Use of the conditioning analysis (Section 3.2) as a design rule for the cross-track baselines of a formation.
- Computational cost: block-wise or GPU implementation of the model and its adjoint for full scenes.

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